Firm Heterogeneity, Misallocation, and Trade

by

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CES 25-33 **May 2025**

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Abstract

To what extent do domestic distortions influence the gains from trade? Using data from Chinese manufacturing surveys and U.S. census records, I document two novel stylized facts: (1) Larger producers in China exhibit lower revenue productivity, whereas larger producers in the U.S. exhibit higher revenue productivity. (2) Larger exporters in China exhibit lower export intensity, whereas larger exporters in the U.S. exhibit higher export intensity. A model of heterogeneous producers shows that only the U.S. patterns are consistent with an efficient allocation. To reconcile the observed patterns in China, I introduce producer- and destination-specific subsidies and estimate the model without imposing functional form assumptions on the joint distribution of productivity and subsidy rates. Accounting for distortions in China leads to substantially smaller estimated gains from trade.

JEL Classification: F14, F12, D61, O17

* Any views expressed are those of the author and not those of the U.S. Census Bureau. The Census Bureau has reviewed this data product to ensure appropriate access, use, and disclosure avoidance protection of the confidential source data used to produce this product. This research was performed at a Federal Statistical Research Data Center under FSRDC Project Number 2698. (CBDRB-FY23-P2698-R10475)

1 Introduction

For some developing countries, trade liberalization acts as a catalyst for economic development and growth. For many others, however, it fails to deliver the anticipated gains. Why do different developing countries reap different benefits from international trade?

One potential explanation lies in market distortions causing suboptimal allocation of production factors across producers. Economists have long recognized that domestic distortions modulate the gains from trade (GFT), which can either alleviate or exacerbate existing inefficiencies (Bhagwati & Ramaswami, 1963). This theoretical ambiguity invites empirical analysis, yet data on relevant sources of distortions, such as political connections, are rarely available. Consequently, the impact of market distortions on the GFT remains underexplored despite substantial policy implications (Atkin & Khandelwal, 2020).

In this paper, I propose a method to infer the presence of producer-level distortions using administrative data and develop an estimation strategy to quantify the GFT while accounting for these distortions. I apply these strategies to Chinese manufacturing survey data from 2000 and U.S. manufacturing census data from 2002. The empirical analysis reveals significant producer-level misallocation in China, which quantitatively matters in assessing the GFT. A model that accounts for this misallocation predicts considerably smaller GFT than the standard model that assumes an efficient market.

I argue that the relationships between revenue productivity, export intensity, and producer size are informative in revealing misallocation. To develop this argument, I establish a set of new stylized facts in Section 2 that highlight stark contrasts between the Chinese and U.S. manufacturing sectors.

First, in China, average revenue productivity decreases with producer size within an industry, while in the U.S., it increases. This pattern holds even when the sample is limited to non-exporting producers, indicating that it is not driven by the previously documented observation that Chinese exporters exhibit lower revenue productivity than their non-exporting counterparts (D. Lu, 2010; J. Lu et al., 2010). Furthermore, the negative relationship observed in China is present among private firms as well as state-owned enterprises (SOEs), suggesting that this pattern cannot be explained solely by differences between the private and state sectors (Jurzyk, 2021).

Second, in China, average export intensity decreases with exporter size within an industry, while in the U.S., it again increases. These patterns persist when pure exporters are excluded and thus cannot be attributed to China's processing trade regime, which results in small exporters with

high export intensity (Defever & Riaño, 2017; Manova & Yu, 2016). This finding is especially surprising given previous studies reporting that larger exporters in China tend to export to a greater number of destinations (Manova & Zhang, 2012; Wang & Yu, 2012).

These patterns remain robust when accounting for differences in sampling rules and measurement errors across countries. The patterns in China also hold in later years and in nearly all industries within the manufacturing sector.

To establish that the patterns observed in the Chinese manufacturing sector indicate producer-level misallocation, I consider the social planner's problem and the resulting allocation in Section 3. In this model, similar to Melitz (2003), consumers enjoy variety, each producer is endowed with a blueprint for a distinct variety and an idiosyncratic productivity, and production and shipment to a given destination market incur fixed overhead and ad valorem shipping costs.

The model predicts that under the planner's allocation, regardless of the shape of the productivity distribution, revenue productivity is non-decreasing with size among non-exporters, and export intensity is non-decreasing with size among exporters. Thus, the patterns observed in the Chinese data contradict the model-predicted patterns under an optimal allocation. Furthermore, these predictions hold across a broad set of model specifications, including cases with non-CES utility and producer-specific costs.

In Section 4, I show that producer-specific subsidies on domestic sales explain the patterns observed in the Chinese data. These producer-specific subsidy rates capture the discrepancy between each producer's market-equilibrium size and its optimal size based on their productivity. A useful interpretation of the idiosyncratic subsidy rate is as a formalization of political connections that confer advantages such as direct subsidies (Alonso et al., 2022), tax benefits (W. Wu et al., 2012), better credit access (Gao et al., 2023), and lower interest rates (Harrison et al., 2019). The contingency on domestic sales, which reflects the World Trade Organization's prohibition of export subsidies, is relaxed in the quantitative model to allow different rates for domestic and export sales and flexibly capture different types of misallocation.

Intuitively, a higher subsidy rate incentivizes a producer to employ more labor than it would otherwise. At the same time, the producer charges a lower price to sell the extra output. Since the subsidy does not affect physical productivity, the lower price translates to lower revenue productivity. Furthermore, because subsidies are contingent on domestic sales, exporters that have grown large due to high subsidy rates exhibit lower export intensity. This creates a negative relationship between export intensity and size, conditional on the set of export destinations, while maintaining a positive relationship between the number of export destinations and size.

Section 5 examines the quantitative importance of accounting for producer-level misallocation

in predicting the effects of trade liberalization. I estimate and calibrate two versions of the model to the same Chinese dataset and compare the GFT predicted by each model. In the baseline version, producers vary only in productivity, which, together with the CES utility assumption, leads to an efficient allocation under market equilibrium. In the full version of the model, producers are additionally endowed with destination-specific subsidy rates. These subsidy rates lead to a suboptimal allocation of labor both across producers and, within each producer, across domestic and foreign destinations.

Due to limited understanding of how political connections and other sources of market distortions are determined for each producer, I treat the subsidy distribution as exogenous and do not model the underlying data-generating process. To address this limitation, I propose an estimation strategy that avoids imposing a functional form assumption on the joint distribution of productivity and subsidy rates across producers.

The two versions of the model predict substantially different GFT. A percentage reduction in exporting costs and import prices leads to a half-percent increase in aggregate real consumption under the baseline model, but less than a third-percent increase under the full model. This is despite both models predicting nearly the same increase in export volume in response to trade liberalization. Therefore, for developing countries, GFT estimation based on trade elasticity alone can be subject to significant specification error. A closer examination suggests that, for China, the lower estimated GFT arises from dampened extensive margin adjustment, resulting in a lower productivity gain from trade-induced reallocation.

This paper contributes to several strands of literature, including research on producer-level misallocation in China. Hsieh and Klenow (2009) document substantially greater dispersion of revenue productivity in China compared to the U.S. Because their model predicts that revenue productivity is equalized across producers under an optimal allocation, the observed dispersion in China implies productivity loss due to misallocation. However, in the setting with overhead fixed costs and international trade, this prediction does not hold. By examining the joint distribution of revenue productivity and size, rather than only the marginal distribution of each, I provide a more robust approach to detecting producer-level misallocation, revealing qualitatively different patterns between China and the U.S.

Khandelwal et al. (2013) identify misallocation of export quotas in China by observing that productivity growth following the removal of quotas was driven by entrants rather than incumbents, which contradicts the predictions of efficient, auction-based allocations. In contrast, I examine cross-sectional patterns and focus on misallocation among non-exporters and exporters, rather than between exporters and potential exporters. Nonetheless, my quantitative model predicts

and captures misallocation along the selection margin identified in their study.

More broadly, trade economists have noted differences between exporters in China and the U.S. Specifically, productivity premium for exporters found in the U.S. and other countries (Wagner, 2007) is absent in China (D. Lu, 2010). My empirical findings focus on patterns *among* exporters and non-exporters. This shift in focus is motivated by the fact that the exporter premium based on *revenue* productivity depends on the underlying distribution of true marginal productivity (Schröder & Sørensen, 2012) and, in theory, can be negative under efficient allocations. Consistent with this, Li et al. (2020) find a positive exporter premium in China when considering *physical* productivity. My model estimation, which uncovers the underlying distribution of marginal productivity and subsidy rates for each observed firm, similarly reveals a positive productivity premium for exporters

The theoretical part of this paper builds on Dhingra and Morrow (2019), which extends Dixit and Stiglitz (1977) by characterizing the efficient allocation under monopolistic competition with heterogeneous producers. These papers identify the conditions on consumer preferences under which the market equilibrium coincides with the social planner's allocation. In contrast, I focus on the observable patterns under an efficient allocation, extending the model to a multi-country setting to make predictions about export intensity.

The paper also contributes to the literature that quantifies the gains from trade liberalization. Two papers, Bai et al. (2024) and Adão et al. (2020), are especially relevant. Like Bai et al. (2024), I use Chinese data and consider a setting with producer-level misallocation. However, my approach does not impose functional form assumptions on the productivity and subsidy distributions, addressing challenges they highlight: unobserved destination-specific labor for each producer and unobserved potential producers. I show that destination-specific labor use can be inferred from the sales ratio in a model-consistent way, and that the issue of potential but inactive producers can be resolved through a careful selection of counterfactual scenarios.

Adão et al. (2020) also provide an estimation strategy that does not impose functional form assumptions on the distribution of producer characteristics, although in a setting without market distortions. They estimate the relevant elasticities using macro data by assuming common elasticity functions across sample economies. In the context of misallocation, however, this assumption is more challenging to justify since the nature of market distortions likely varies across countries and time periods, My approach addresses this issue by directly estimating the underlying functions from producer-level data.

2 Stylized facts

2.1 Data

To analyze the Chinese manufacturing sector, I use firm-level industrial survey data collected by the National Bureau of Statistics of China.[1] For the main analysis, I focus on the year 2000, capturing a snapshot of China's manufacturing sector just before its accession to the World Trade Organization (WTO). The survey includes all SOEs and non-state-owned firms with sales above 5 million Renminbi, or approximately 600 000 US dollars. Each firm is classified under a 4-digit industry code based on China's National Standards code GB/T 4754.

The analysis sample is restricted to firms in one of 446 manufacturing industries with positive employment and sales. I assign each firm to one of three ownership statuses based on its registration type: domestic private, state-owned, and foreign. In 2000, approximately 19 % of firms were state-owned, and 19 % were foreign-owned.

The U.S. data come from the Census of Manufactures conducted by the Bureau of the Census. I focus on the year 2002, the census year closest to 2000. This dataset includes all manufacturing establishments with one or more paid employees. As with the Chinese dataset, I exclude observations with non-positive sales. Each establishment is classified under a 6-digit NAICS industry code based on its primary product. Throughout the paper, I use the term producer to refer to firms in the Chinese national survey and establishments in the U.S. Census.

Within each dataset, I partition producers into two groups: non-exporters, defined as those reporting zero export sales, and exporters. A concern when comparing exporters between the two countries is the prevalence of processing trade in China. Under the processing trade regime, a firm imports materials and re-exports the processed or assembled products. Previous research has documented that exporters primarily engaged in processing trade exhibit lower productivity and size compared to ordinary exporters and even non-exporters. While the survey data do not directly indicate whether an exporter engages in processing trade, I exclude firms with an export intensity greater than 90 % to ensure that the documented patterns are not driven purely by processing exporters.[2] For comparability, I apply the same 90 % export intensity threshold to the U.S. data.

Three producer-level outcomes are of interest: size, revenue productivity, and export intensity. I use total wage payments as the primary measure of producer size, capturing quality-adjusted labor use. As an alternative measure of input size, I take the weighted geometric mean of wage

[1] Brandt et al. (2014) provide a detailed description of the survey.

[2] Dai et al. (2016) find that more than half of processing firms export all of their products, compared to 14 % among ordinary exporters.

payments and the book value of fixed assets, with weights calculated at the aggregate industry level.[3] The primary measure of revenue productivity is the ratio of total sales to total wage payments, with alternative measures using value-added as the output. Lastly, export intensity is measured as the export share of total shipment value.

Table 1: Summary statistics

Variable	Chinese manufacturing, 2000		U.S. manufacturing, 2002	
	Non-exporters	Exporters	Non-exporters	Exporters
Log sales	9.28	10.48	2.86	4.05
	(1.50)	(1.42)	(0.80)	(0.70)
Log wages	6.55	7.77	2.30	3.39
	(1.32)	(1.41)	(0.71)	(0.59)
Log sales-to-wage	2.73	2.71	0.56	0.66
	(1.16)	(0.94)	(0.27)	(0.28)
Export share of sales		0.36		0.12
		(0.29)		(0.16)
Observations	104132	21631	310000	39000

Note: The table reports the mean and the standard deviation in parentheses of each variable for each subgroup of the producers. The exporters exclude producers whose export share of sales is greater than 90 %. The number of observations for the U.S. is rounded to the nearest thousand.

Table 1 reports the mean and standard deviation of the main variables for each group of producers and country. These summary statistics reveal differences in exporter premia between the countries. In both countries, exporters are, on average, larger than non-exporters in both sales and wages. However, the average sales-to-wage ratio is roughly the same for exporters and non-exporters in China, while it is higher for exporters in the U.S.

2.2 Revenue productivity

Figure 1 illustrates the relationship between producer size and revenue productivity using locally smoothed polynomial plots. Producer size is shown on the horizontal axis, measured by total wages, while revenue productivity is shown on the vertical axis, measured by the sales-to-wage ratio.

Subfigure 1(a) presents this relationship for the Chinese manufacturing sector in 2000, while

[3]For each firm i in sector s, the (log) factor bundle is $\alpha_s \log(L_i) + (1-\alpha_s)\log(K_i)$, where L_i is total wage payment, K_i is net fixed assets, and $\alpha_s = \left(\sum_{i\in s} L_i\right) / \left(\sum_{i\in s} VA_i\right)$. For China, I use the 3-digit GB/T industry code as the sector classification; for the U.S., I use the 4-digit NAICS code.

Subfigure 1(b) shows for the U.S. in 2002. Only non-exporting producers are included for both countries.[4] Both variables are demeaned at the industry level to remove cross-industry variation. The local polynomial plots are overlaid with a kernel density plot of the total wage distribution.[5]

The figure highlights a stark difference in the size-productivity relationship between the two countries. In China, larger firms tend to exhibit lower revenue productivity than smaller firms, whereas in the U.S., the opposite pattern is observed.

A potential explanation for this discrepancy is the significant share of SOEs in China, which tend to be larger but less productive than private firms (Brandt et al., 2013). Figure 2 shows the same size-productivity relationship in China as in Figure 1(a), but separately for SOEs and non-SOEs. The plot confirms that SOEs tend to have lower revenue productivity than similarly sized private firms. More importantly, it also reveals that the negative relationship persists within both non-SOEs and SOEs. Thus, SOEs alone do not explain the observed discrepancy between China and the U.S. — large private firms in China tend to be less productive than smaller private firms, and large SOEs tend to be less productive than smaller SOEs.

Table 2 presents the regression estimates of revenue productivity on producer size, with results for the Chinese manufacturing sector in Panel A and for the U.S. manufacturing sector in Panel B. In all columns, the dependent variable is revenue productivity measured by log sales-to-wage, and the sample includes only non-exporting producers.

Column (1) presents the OLS regression of log sales-to-wage on log wages. The precisely estimated coefficients indicate that a 1 % increase in total wages is associated with a 0.24 % decrease in revenue productivity for Chinese firms, whereas it corresponds to a 0.07 % increase for U.S. establishments.

Column (2) adds industry fixed effects, using the 4-digit Chinese industry classification code in Panel A and the 4-digit NAICS code in Panel B. The qualitative difference between the countries remains with the inclusion of industry fixed effects, reflecting the patterns shown in Figure 1. Column (3) further adds region fixed effects. A region is defined as a province in China (Panel A) and a state in the U.S. (Panel B). These fixed effects account for potential reporting discrepancies across regions, which may be especially problematic in the Chinese dataset. Overall, the patterns remain unchanged, with a slightly higher coefficient magnitude observed in China.

Lastly, Column (4) includes indicators for SOE and foreign ownership as additional regressors for the Chinese dataset. SOEs tend to have lower revenue productivity, consistent with Figure 2, while foreign-owned firms tend to have higher revenue productivity. Furthermore, the rev-

[4]The patterns remain when exporters are included, as suggested by the summary statistics in Table 1.
[5]In accordance with Census disclosure rules, the U.S. plots are truncated at the top and bottom 5 %.

Table 2: Revenue productivity and producer size

	(1)	(2)	(3)	(4)
	Panel A. Chinese manufacturing, 2000			
Log wage	-0.244	-0.244	-0.257	-0.280
	(0.003)	(0.003)	(0.003)	(0.003)
𝟙(SOE)				-0.883
				(0.009)
𝟙(Foreign-owned)				0.144
				(0.010)
Observations	104132	104126	104126	104126
R^2	0.076	0.192	0.252	0.341
	Panel B. US manufacturing, 2002			
Log wages	0.065	0.020	0.020	
	(0.001)	(0.001)	(0.001)	
Observations	310000	310000	310000	
R^2	0.031	0.386	0.386	
Industry FE	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes

Note: The dependent variable is the log ratio of total sales to total wages. The unit of observation is a firm in Panel A and an establishment in Panel B. The number of observations in Panel B is rounded. Industry is a 4-digit Chinese industry classification code in Panel A, and 4-digit NAICS in Panel B. Region is a Chinese province in Panel A, and a U.S. state in Panel B. Robust standard errors are reported in the parentheses.

enue productivity differences across ownership types do not explain the negative relationship between revenue productivity and firm size in China, with the coefficient on firm size remaining significantly negative at -0.28.

Table 3 provides additional regression results to verify that the baseline results for China are not driven by data idiosyncrasies. One concern with the Chinese survey is the above-size sampling. Because private firms are surveyed only if their revenue is above a certain threshold, firms with small employment and wage payments included in the survey will have higher revenue productivity than other similarly sized firms.

This concern about selection bias is mitigated by two observations. First, as Figure 1(a) shows, the negative relationship is not solely driven by small firms but holds across all firm sizes. Second, because all SOEs are included in the survey, they are not subject to the revenue threshold issue.

Table 3: Revenue productivity and producer size in China: robustness

	SOE only	IV	VA	Input mixture	All firms
	(1)	(2)	(3)	(4)	(5)
Log size	-0.127	-0.193	-0.273	-0.529	-0.241
	(0.006)	(0.004)	(0.004)	(0.003)	(0.003)
Industry FE	Yes	Yes	Yes	Yes	Yes
Province FE	Yes	Yes	Yes	Yes	Yes
Observations	24976	104126	93201	92781	140767
R^2	0.225	0.081	0.192	0.344	0.245

Note: The dependent variable is the log ratio of total sales to total wages in Columns (1), (2), and (5). It is the log ratio of value-added to total wages in Column (3) and log ratio of value-added to wage-capital mixture in Column (4). The main regressor, log size, is log of wage-capital mixture in Column (4) and log wages in all other columns. The unit of observation is a firm. The sample is restricted to non-exporting firms in Columns (1) to (4) but includes all firms in Column (5). Column (1) further restricts the sample to SOEs. Column (2) uses total employment as an instrument for the total wages. Robust standard errors are reported in the parentheses.

As Figure 2 shows, the negative relationship holds even among SOEs. This is confirmed in Table 3 Column (1), which restricts the sample to SOEs.

Another concern is the measurement error associated with total wages. Because total wages appear both as an independent variable and in the denominator of the dependent variable, measurement errors associated with it lead to a negative bias. To address this concern, I use total employment as an instrumental variable. This strategy addresses classical measurement errors that are independent across variables. The corresponding coefficient estimate, reported in Column (2), shrinks to -0.19, consistent with the OLS estimate being negatively biased due to measurement error. However, it remains significantly negative, affirming that the qualitative patterns are not driven by measurement errors alone.

Columns (3) and (4) use alternative measures of firms' outputs and inputs. Instead of total sales, Column (3) uses value-added as the output variable. The coefficient estimate remains nearly unchanged, suggesting that within industry-region, the value-added share of revenue variation is limited across firms. Column (4) additionally considers an alternative input measure, using a Cobb-Douglas combination of total wages and the book value of capital. The resulting coefficient remains negative, but with a larger magnitude at -0.53.

Lastly, for the purpose of fully documenting the patterns, Column (5) incorporates all firms in the sample, including exporters. The results confirm that the negative relationship between firm size and revenue productivity in China holds when exporters are included.

2.3 Export intensity

Figure 3 shows the relationship between producer size, measured by total wages on the horizontal axis, and export intensity, measured by the share of exports in total sales on the vertical axis, using local polynomial smoothing. As in Figure 1, both variables are demeaned at the industry level.

The plots reveal another clear qualitative difference between the two countries. In China, larger exporters exhibit lower export intensity, while in the U.S., larger exporters tend to exhibit higher intensity.

One possible explanation for this difference is the prevalence of processing exporters in China, which exhibit high export intensity despite being smaller than ordinary exporters. The exclusion of exporters with above-threshold intensity may not fully assuage this concern, given the ad hoc choice of the 90 % threshold. However, Figure 3 shows a negative relationship between size and export intensity in China even among medium-sized and large exporters, suggesting that the pattern is not solely driven by any remaining processing exporters included in the sample. Such small but high-intensity exporters are rarer but still present in the U.S. Census data, partly explaining the negative relationship observed among smaller exporters in the U.S.

Table 4 shows the coefficient estimates from a linear regression with log export intensity as the dependent variable and log wages as the main independent variable. Consistent with Figure 3, the main coefficient estimate is negative for China but positive for the U.S., with or without industry and region fixed effects.

Column (4) reports the results including ownership type fixed effects for China and reveals that SOEs tend to have lower export intensity, while foreign-owned firms have higher export intensity. The coefficient sign on log wages remains unchanged, indicating that the association is not solely explained by SOE ownership.

2.4 Discussion

I have shown that revenue productivity and export intensity relate to producer size differently in China and the U.S. In China, both tend to decline with size, while in the U.S., both tend to increase.

Previous studies suggest that the patterns found in the U.S. are the norm, at least among the developed countries. Bartelsman et al. (2013) measure the within-industry covariance between revenue-based productivity and size for seven European countries and the U.S., finding only one country, Romania, that exhibits negative covariance.

Table 4: Export intensity and producer size

	(1)	(2)	(3)	(4)
	Panel A. Chinese manufacturing, 2000			
Log wage	-0.247	-0.146	-0.136	-0.127
	(0.008)	(0.008)	(0.008)	(0.008)
𝟙(SOE)				-0.194
				(0.033)
𝟙(Foreign-owned)				0.292
				(0.023)
Observations	21631	21616	21616	21616
R^2	0.051	0.223	0.236	0.246
	Panel B. US manufacturing, 2002			
Log wages	0.072	0.064	0.074	
	(0.006)	(0.007)	(0.007)	
Observations	39000	39000	39000	
R^2	0.003	0.144	0.149	
Industry FE	No	Yes	Yes	Yes
Region FE	No	No	Yes	Yes

Note: The dependent variable is the log ratio of export sales to total sales. The unit of observation is a firm in Panel A and an establishment in Panel B. The number of observations in Panel B is rounded. Industry is a 4-digit Chinese industry classification code in Panel A, and 4-digit NAICS in Panel B. Region is a Chinese province in Panel A, and a U.S. state in Panel B. Robust standard errors are reported in the parentheses.

Similarly, average export intensity has been found to increase with firm size in Italy (Majocchi et al., 2005), the Netherlands (Verwaal & Donkers, 2002), and Spain (Hernández, 2019). Exceptionally, Patibandla (1995) find that export intensity tends to fall with firm size in India and attribute this to "pervasive market imperfections and sub-optimal economic institutions" as possible explanations.

Because the patterns observed in China are more unusual, I provide additional robustness checks for the Chinese data. First, I examine whether the patterns hold in years other than 2000. Appendix Figure A.1 shows results for the years 2002, 2004, and 2006. The relationship between revenue productivity and size remains stable over the years. The export intensity pattern, however, shows more variation over time, and by 2006, the average export intensity rises with size among small to medium-sized firms. Nonetheless, the negative relationship persists

among the larger exporters, in contrast to the U.S.

Second, I verify that the patterns are not driven by a small number of large industries within manufacturing. Specifically, I regress the dependent variable, either the log sales-to-wage ratio or export intensity, on log total wages interacted with 2-digit sector indicators, while also including industry and region fixed effects. Appendix Figure A.2 plots the coefficient estimates for each sector interaction term. The figure confirms that the negative relationship for both revenue productivity and export intensity holds within nearly all 2-digit sectors, indicating that the aggregate result is not driven by a few large sectors.

3 Patterns under optimal allocation

In this section, I provide a theoretical framework to interpret the differences in producer-level patterns observed between China and the U.S.

I consider the social planner's problem of maximizing the representative consumer's utility by allocating a national endowment of labor across producers with heterogeneous productivity. As in Dhingra and Morrow (2019), the planner balances quantity and variety to determine the optimal allocation. I extend this setting by requiring exports to meet a utility quota associated with each destination country. Thus, the planner not only allocates labor across producers, but also determines how labor is divided within each producer across potential destination countries.

The key result is that, under the planner's allocation, the sales-to-employment ratio among non-exporting producers and the export intensity among exporters are non-decreasing with employment. Therefore, the patterns observed in the Chinese manufacturing sector indicate market distortions.

3.1 Model setting

The world consists of $N > 1$ countries. For notational convenience, I describe the consumer and producers of country 1. Other countries have analogous preferences and production technologies.

The country's representative consumer has a constant elasticity of substitution (CES) utility function

$$U := \int_{\omega \in \Omega} q(\omega)^{\rho} \, \mathrm{d}\omega \,, \tag{1}$$

where ω indexes a product variety, $\Omega \subseteq \mathbb{R}$ is the set of all varieties available in the country, and

$q(\omega)$ is the quantity of variety ω consumed. The exponent $\rho \in (0,1)$ measures the consumer's love of variety and determines the elasticity of substitution $\sigma = (1-\rho)^{-1} > 1$.

The representative consumer is endowed with $\bar{L} > 0$ units of labor, the only factor of production. The country has a unit mass of producers, indexed by $i \in [0,1]$, each endowed with a blueprint for a distinct variety. As such, no two producers in the world produce the same variety. Each producer can produce and ship its variety to different countries $j \in \{1,\ldots,N\}$. Producer i requires $l_j(q;i)$ units of labor to produce and ship q units to country j.

Each producer i can convert a unit of labor into $z(i) > 0$ units of its variety, where the distribution of $z(i)$ across producers is assumed to be absolutely continuous. Delivering one unit to destination j requires shipping $t_j \geq 1$ units. I assume $t_1 = 1$, so that delivery to the domestic market incurs no loss. Finally, shipping any positive amount of variety to destination j incurs a destination-specific fixed overhead cost $f_j > 0$ units of labor. The presence of fixed costs implies increasing returns to scale within each destination.

In summary, the labor cost is described by

$$l_j(q;i) := f_j \mathbb{1}(q > 0) + \frac{t_j q}{z(i)}, \tag{2}$$

where $\mathbb{1}(P)$ is an indicator function that takes the value of one if the predicate P is true and zero otherwise.

The production cost is additively separable across destinations, so the total labor required to produce and ship q_j units of variety to each destination j is simply the sum $l(q_1,\ldots,q_N;i) = \sum_{j=1}^{N} l_j(q_j;i)$.

3.2 Planner's problem

I consider the problem of the global social planner who maximizes the weighted sum of utilities across countries. Labor is immobile across countries, so the planner allocates each country's labor among the producers within that country. I refer to the resulting allocation as efficient or optimal.

Because the utility of each consumer is additively separable across varieties from different origins, and the production cost of each producer is additively separable across destinations, the global planning problem can be divided into a set of national planning problems. The global planner only needs to inform the national planners how much of each country's labor should be allocated toward shipments to country j.

Country 1's national planner problem for destination j is

$$\begin{aligned} \max_{\{l_j(i)\}_{i\in[0,1]}} \quad & \int_0^1 q_j(i)^\rho \,\mathrm{d}i \\ \text{s.t.} \quad & \int_0^1 l_j(i)\,\mathrm{d}i = \bar{L}_j, \\ & l_j(i) = f_j \mathbb{1}_{q_j(i)>0} + \frac{t_j q_j(i)}{z(i)} \quad \forall j \in \{1,\dots,N\}\, \forall i \in [0,1]. \end{aligned} \tag{3}$$

where ρ, f_j, t_j, $\{z(i)\}_{i\in[0,1]}$, and $\bar{L}_j \geq 0$ are given. The total labor allocated for exports to j, $\bar{L}_j$, satisfies $\sum_j \bar{L}_j = \bar{L}$ and is determined by the global planner, partly based on the utility weights across countries.[6]

Lemma 1 characterizes the solution to problem (3).

Lemma 1. Let $q_j(i)$ denote the quantity of variety produced and shipped to market j by producer i under the optimal allocation, i.e. solution to (3). Let $A_j := \{i \in [0,1] \mid q_j(i) > 0\}$ denote the set of producers active in market j. Then

$$q_j(i) = \mu_j \tilde{z}_j(i)^{\frac{1}{\rho}}, \quad \tilde{z}_j(i) := \left(\frac{z(i)}{t_j}\right)^{\frac{\rho}{1-\rho}}, \quad \text{for all } i \in A_j \tag{4}$$

where $\mu_j > 0$ is common across producers. Furthermore, there exists $a_j \geq 0$ such that

$$i \in A_j \iff z(i) \geq a_j.$$

Proof. Appendix B. □

[6] Specifically, the global planner solves

$$\begin{aligned} \max_{\{l_{od}(i)\}_{o,d\in\{1,\dots,N\},i\in[0,1]}} \quad & \sum_{d=1}^N W_d \left(\sum_{o=1}^N \left(\int_0^1 q_{od}(i)^\rho \,\mathrm{d}i \right) \right) \\ \text{s.t.} \quad & \sum_{d=1}^N \left(\int_0^1 l_{od}(i)\,\mathrm{d}i \right) = \bar{L}_o \quad \forall o \in \{1,\dots,N\}, \\ & l_{od}(i) = f_{od}\mathbb{1}_{q_{od}(i)>0} + \frac{t_{od} q_{od}(i)}{z_o(i)} \quad \forall o,d \in \{1,\dots,N\}\, \forall i \in [0,1], \end{aligned}$$

where $W_d > 0$ is the utility weight given to country d, $q_{od}(i)$ is the quantity of variety produced by producer i in country o and consumed in country d, f_{od} and t_{od} are fixed and variable costs associated with sales from country o to country d, $z_o(i)$ is the productivity of producer i in country o, and $\bar{L}_o$ is the labor endowment in country o. Country 1 planner takes $\bar{L}_j = \int_0^1 l_{1j}^*(i)\,\mathrm{d}i$ as given, where $l_{od}^*(i)$ is the global planner's solution.

3.3 Expected data patterns under optimal allocation

In this subsection, I consider the observable implications of the planner's solution. Since the data contain values rather than quantities of shipments, assumptions about price determination are necessary. A natural assumption is that observed shipment values reflect prices determined by the demand derived from the utility function (1):

$$p_j(q) = D_j q^{\rho-1}, \tag{5}$$

where $D_j > 0$ is the demand shifter for destination j, common across all varieties.

I focus on the distributions of $r(i) := \sum_{j=1}^{N} p_j(q_j(i))q_j(i)$ and $l(i) := \sum_{j=1}^{N} l_j(q_j(i);i)$. These model variables correspond to each producer's total shipment value and employment (or total wages, normalizing wage rate to unity) in the data, respectively.

Lemma 2 characterizes the sales and employment associated with each destination under the optimal allocation.

Lemma 2. Let $q_j^*(i)$ denote the value of $q_j(i)$ that solves the planner's problem. Then for all $i \in A_j$,

$$r_j(i) := p_j(q_j^*(i))q_j^*(i) = D_j \mu_j^{\rho} \tilde{z}_j(i) \tag{6}$$

and

$$l_j(q_j(i);i) = f_j + \mu_j \tilde{z}_j(i). \tag{7}$$

Proof. The results follow directly from (2), (4), and (5). □

One difficulty in testing Lemma 2 is that we only observe total employment, $l(i)$, without information on how labor is split between domestic sales and exports. To address this, I consider the implications of Lemma 2 separately for non-exporting producers and exporters. For non-exporters, the observed total employment necessarily corresponds to the employment for domestic sales, $l(i) = l_1(i)$, allowing inference on the relationship between sales and employment. For exporters, I focus on the relationship between export and domestic sales.

Claim 1. Under optimal allocation, the shipment value to total employment ratio is non-decreasing in total employment among non-exporting producers.

Proof. Appendix B. □

The ratio of output quantity to production employment corresponds to the producer's productivity. Due to the CES demand, the price that clears the optimal quantity is inversely proportional to

productivity, meaning the ratio of sales to production employment is equalized across producers. Consequently, larger producers that amortize the fixed overhead labor exhibit a higher ratio of sales to total employment.

A few remarks are in order. First, the claim does not generalize to all producers. As noted by Schröder and Sørensen (2012), conditional on productivity $z(i)$, the sales-to-employment ratio decreases when producer i exports due to the associated fixed overhead cost f_j. Thus, the monotonicity result of Claim 1 does not hold across all producers, and depending on the productivity distribution, the average sales-to-employment ratio of exporters can be lower than that of non-exporters under an optimal allocation.

Second, the claim should be distinguished from the use of size and productivity covariance as a measure of allocative efficiency, as in Olley and Pakes (1996). Higher covariance of size and productivity indicates greater allocative efficiency only if the objective is solely to maximize total output quantity without regard to product variety, and if productivity captures output quantity per input rather than sales value per input. When consumers exhibit a love of variety and measured productivity is based on sales value, the magnitude of the size-productivity covariance does not reflect allocative efficiency. Indeed, in the special case where fixed overhead costs are absent ($f_1 = 0$), the sales-to-employment ratio is equalized across all firms under an optimal allocation (Hsieh & Klenow, 2009), resulting in zero covariance. Claim 1 does suggest, however, that the *sign* of the correlation, when restricted to non-exporting producers, can be used as a test for optimal allocation without making assumptions about the productivity distribution or other parameter values.

Claim 2 provides a testable implication of Lemma 2 for exporting producers.

Claim 2. Suppose $a_1 < a_j$ for all $j \neq 1$. Under the optimal allocation, the export share of total shipments is non-decreasing in total employment among exporting firms.

Proof. Appendix B. □

The result of Claim 2 is driven by the extensive margin. The planner assigns only a subset of producers to export due to the associated overhead costs. Because the assignment is based on the productivity, more productive producers export to a greater number of destinations.

On the intensive margin, due to isoelastic demand and common trade costs, export intensity is equalized conditional on the set of destinations. However, in the next subsection, I show that under more general and empirically relevant settings, export intensity also increases with size on the intensive margin.

3.4 Extensions

The fact that the expected patterns under optimal allocation, as described in Claims 1 and 2, fail to hold in the Chinese manufacturing sector suggests suboptimal allocation. This conclusion is valid, however, only if the model's preference and technology assumptions hold. In this subsection, I relax these assumptions and consider a more general specification of preferences and producer-specific trade costs. This generalization supports the robustness of the claims, which continue to hold under empirically plausible conditions. Proofs for the claims in this subsection are provided in Appendix B.

3.4.1 Variable elasticity of substitution

The assumption that utility exhibits constant elasticity of substitution (CES) can be relaxed to allow for variable elasticity of substitution (VES):

$$U := \int_{\omega \in \Omega} u(q(\omega))\, \mathrm{d}\omega\,, \tag{8}$$

where the inner function u satisfies the Inada conditions. The CES case is nested under $u(q) = q^{\rho}$.

In Appendix B, I show that Claim 1 still holds. The social planner chooses output quantity of each producer by balancing the marginal benefit and cost of a variety. Consequently, the price of each variety remains proportional to its marginal cost under the planner allocation, as in the CES case.

Claim 2 also holds if $\beta(q_j) \leq \beta(q_1)$ for foreign destinations $j > 1$, where $\beta(q) := -\frac{q u''(q)}{u'(q)}$ is the elasticity of marginal utility. Assuming that the utility weights on foreign countries are not too high so that $q_j(i) \leq q_1(i)$, this inequality holds if the elasticity of marginal utility is non-decreasing. As discussed by Dhingra and Morrow (2019), empirical literature largely supports increasing elasticity of marginal utility. Therefore, the empirically relevant case of the VES generalization strengthens Claim 2 by predicting a positive relationship between export intensity and size on both the intensive and extensive margins.

3.4.2 Producer-specific variable trade costs

Suppose producers vary not only in productivity but also in their variable trade costs associated with each foreign destination. For example, a producer may incur lower variable trade costs if it is located closer to a port or produces a variety more amenable to international shipping. Specifically, assume that the idiosyncratic trade cost of producer i can be decomposed into a destination-specific cost common across producers and a producer-specific cost common across

destinations, so that $t_j(i) := t_j \tilde{t}(i)$ for each $j = 2, \ldots, N$. Each producer is now characterized by both its productivity and trade cost, $(z(i), \tilde{t}(i))$.

This extension does not affect Claim 1, which characterizes only non-exporting producers, since the revenue productivity of non-exporting producers is invariant to trade costs.

Consider Claim 2, which relates export intensity to the size of exporting producers. The extensive margin result remains unaffected — larger producers still export to a greater number of destinations under the optimal allocation.

On the other hand, heterogeneity in trade costs introduces variation on the intensive margin. By Lemma 2, for a producer i active in destination j, the sales ratio $r_j(i)/r_1(i)$ decreases with $t_j(i)$. At the same time, conditional on productivity $z(i)$, a producer with higher trade cost $t_j(i)$ is allocated less labor. This observation suggests that under a "typical" joint distribution of $z(i)$ and $\tilde{t}(i)$, including when these are uncorrelated, export intensity increases in size on the intensive margin as well.

The claim could fail, however, if productivity $z(i)$ and idiosyncratic trade cost $\tilde{t}(i)$ are highly positively correlated. Intuitively, productivity does not affect the optimal export intensity on the intensive margin because it raises the optimal shipment values equiproportionately across destinations. At the same time, productivity increases the optimal employment size. Thus, if large, productive producers idiosyncratically face higher trade costs that dampen export intensity, the average export intensity can decrease with producer size.

It is unclear, however, why more productive producers in Chinese manufacturing, but not in the U.S., would face higher international shipping costs. For instance, Brandt et al. (2013) find that coastal provinces tend to exhibit higher TFP than inland provinces, suggesting a negative rather than positive correlation between productivity and shipping cost in China.

3.4.3 Heterogeneous fixed overhead costs

Suppose now that producers vary in both productivity and fixed overhead costs. Again, I assume fixed cost can be decomposed into destination-specific and producer-specific components, so that $f_j(i) := f_j \tilde{f}(i)$ for $j = 1, \ldots, N$.

When the variance of fixed cost $\tilde{f}(i)$ is sufficiently large relative to that of productivity $z(i)$, both Claims 1 and 2 can fail. To see why, consider the extreme case where all producers share the same productivity level, $z(i) = z$, varying only in overhead costs, $\tilde{f}(i)$. In this setting, conditional on being active, producers with higher idiosyncratic overhead costs are assigned more labor since the optimal allocation of production employment depends only on productivity. At the

same time, high overhead costs lead to lower revenue productivity, which is proportionate to the share of variable cost. This results in a reversal of Claim 1.

The extreme assumption that producers vary only in overhead costs is clearly rejected by the fact that domestic sales vary across producers. Under heterogeneous productivity, the selection effect mitigates the impact of overhead costs on the revenue productivity pattern. Higher overhead costs raise the productivity threshold for a producer to be active. As a result, conditional on being active, producers with larger employment due to higher overhead costs also exhibit higher average productivity. Since productivity increases the sales-to-employment ratio, the selection effect creates a positive relationship between size and sales-to-employment, offsetting the intensive margin effect.

As shown in Appendix B, this intuition manifests in the form of a lower bound on the sales-to-employment ratio, even when the support of the overhead cost distribution is unbounded above. This limited range of sales-to-employment ratio due to selection restricts the intensive margin effect of overhead cost heterogeneity.

Figure 4(a) illustrates this idea using a simulation. Producers draw productivity and fixed overhead costs from a bivariate lognormal distribution, with higher variance in fixed costs to emphasize their impact. The figure plots the domestic sales-to-employment ratio against domestic employment size and reveals a negative relationship across all producers, as higher fixed costs increase employment while lowering productivity.

The selection effect, however, offsets this pattern. In the simulation, the parameters are implicitly chosen so that 80 % of producers are active under the planner's allocation. This imposes a lower-bound on the sales-to-employment ratio, as indicated by the horizontal dash-dot line. The solid curve shows the relationship between revenue productivity and employment when the sample is restricted to active producers. The negative relationship flattens and shows a slight upward trend for larger firms. While the exact shape and direction of this relationship depends on the distributional assumptions and survival threshold, the simulation demonstrates how the selection mechanism helps sustain a positive association between sales-to-employment and employment under heterogeneous fixed costs.

Next, consider the implications of heterogeneous fixed overhead costs on Claim 2. A producer's value of shipments to destination j, conditional on shipping there, is invariant to its overhead cost $f_j(i)$. Therefore, overhead cost heterogeneity affects export intensity only through the extensive margin.

Again, it is informative to consider the extreme case in which producers vary only in overhead costs and not in productivity. The planner is less likely to select producers with high idiosyncratic

costs to export to a given destination. Therefore, high-cost producers export to fewer destinations and exhibit lower export intensity. At the same time, conditional on the set of destinations exported, high-cost producers require more labor to cover the overhead costs. As a result, export intensity becomes negatively associated with size.

This explanation, however, contradicts prior literature's observation that the number of destinations is positively associated with employment size, even for Chinese manufacturers. [7] Therefore, the negative association between export intensity and employment observed in China requires an intensive margin explanation, which heterogeneity in fixed overhead costs fail to provide.

4 Producer-specific subsidies

In light of Claims 1 and 2, the patterns in the Chinese manufacturing data — negative relationships between revenue productivity and export intensity with size — point to market distortions. To capture these distortions, I introduce a model with producer-specific subsidies. These subsidies reflect various sources of inefficiencies, such as political connections, and enable the model to reconcile the observed patterns in China.

Leveraging the result of Dhingra and Morrow (2019) that the monopolistic competition market equilibrium under CES utility achieves an optimal allocation, I shift the focus from the planner's solution to the market equilibrium. The model setting ensures that, in the absence of subsidies, the market equilibrium is efficient. Consequently, introducing subsidies necessarily (weakly) reduces the consumer utility.

The idiosyncratic subsidy rate captures the gap between each producer's optimal and observed size. While such deviations can arise from various factors, scholars have identified political connections as a significant source in China. For example, Milhaupt and Zheng (2015) note that "large firms in China — whether SOEs, POEs, or ambiguous state-private blends — survive and prosper precisely *because* they have fostered connections to state power and have succeeded in obtaining state-generated rents." Similarly, M. Wu (2016) and Pearson et al. (2021) emphasize the pervasive role of the Communist Party in the Chinese economy, even outside of state-owned enterprises.

Producer-specific subsidies in the model formalize the advantages derived from Party-state relationships. These advantages manifest as direct subsidies, tax benefits, or indirect support, such as preferential loans (Firth et al., 2009; Harrison et al., 2019; Song et al., 2011). More broadly,

[7] Using the manufacturing survey linked to customs data, Wang and Yu (2012) find that among Chinese exporters, the number of destination markets is positively associated with employment size, whether or not processing exporters are excluded.

subsidies capture how state connections distort profit-maximization by incentivizing firms to align their behavior with state objectives, such as stabilizing employment to prevent unrest (Wen, 2023).

For simplicity, I consider subsidies only on domestic sales in this section. This assumption aligns with WTO rules that prohibit export-contingent subsidies.[8] When examining the quantitative implications in Section 5, I allow subsidies on export sales to more flexibly capture the observed patterns.

4.1 Setting

I extend the setting introduced in Section 3.1 by allowing each producer i to be endowed with an idiosyncratic subsidy multiplier, $\eta(i) \geq 0$, on domestic sales. Producer i receives a subsidy equal to the product of its domestic sales and $(\eta(i) - 1)$. If $\eta(i) < 1$, the producer pays the absolute value as a tax.

With the subsidy, producer i solves the following profit maximization problem:

$$\max_{\{q_j\}_{j=1}^N} \quad \big(\eta(i) p_1(q_1) q_1 - l_1(q_1; i)\big) + \sum_{j=2}^{N} \big(p_j(q_j) q_j - l_j(q_j; i)\big). \tag{9}$$

Without loss of generality for the results in this section, the wage rate is assumed to be one. The first summand represents the profit from domestic shipments, with revenue adjusted by the producer-specific subsidy multiplier $\eta(i)$. When $\eta(i) \neq 1$, the social and private benefits of domestic production are misaligned, leading to market inefficiency. The remaining summands capture the profits from export shipments, which are not directly affected by $\eta(i)$.

To characterize the solution of the profit maximization problem (9), it is useful to define two producer-specific variables. The domestic *profit potential* of producer i is defined as $h(i) := \eta(i)^{\frac{1}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}}$. At the interior solution, producer i's profit $\pi_1^*(i)$ from the domestic market, inclusive of the subsidy, and the employment used for domestic sales $l_1^*(i)$ are determined by its domestic profit potential:

$$\pi_1^*(i) = (1-\rho)\rho^{\frac{\rho}{1-\rho}} D_1^{\frac{1}{1-\rho}} h(i) - f_1, \quad l_1^*(i) = f_1 + \big(\rho D_1\big)^{\frac{1}{1-\rho}} h(i). \tag{10}$$

The domestic *sales potential* of producer i is defined as $k(i) := \eta(i)^{\frac{\rho}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}}$. The domestic out-

[8] The producer-specific subsidies on domestic sales considered in the model would be "actionable" but not "prohibited" under WTO rules, meaning other member countries can challenge such subsidies through dispute settlement or countervailing measures. As M. Wu (2016) points out, both types of subsidies are subject to legal ambiguities when provided by non-government entities with state connections.

put quantity, price, and shipment value at the interior solution are determined by the producer's sales potential:

$$q_1^*(i) = (\rho D_1)^{\frac{1}{1-\rho}} k(i)^{\frac{1}{\rho}}, \quad p_1^*(i) = \rho^{-1} k(i)^{-\frac{1-\rho}{\rho}}, \quad r_1^*(i) := p_1^*(i) q_1^*(i) = \rho^{\frac{\rho}{1-\rho}} D_1^{\frac{1}{1-\rho}} k(i).$$

In summary, relative to the social optimum, producers with higher subsidy rates produce more, charge lower prices, employ more workers, and enjoy higher profits in the domestic market.

Producer i's decision to be active in the domestic market is also impacted by its subsidy rate. Since profit is increasing in profit potential, there is a threshold value $h^* > 0$ such that a producer makes sales if and only if its profit potential exceeds this threshold:

$$q_1^*(i) > 0 \iff h(i) \geq h^*. \tag{11}$$

4.2 Revenue productivity and size

Under profit maximization, the ratio of domestic sales to the variable cost depends on the subsidy rate. Consequently, this ratio now varies across producers:

$$\frac{p_1^*(i) q_1^*(i)}{l_1^*(i) - f_1} = \frac{1}{\rho \eta(i)}.$$

The sales-to-employment ratio of a non-exporting producer is

$$\frac{r(i)}{l(i)} = \frac{1}{\rho \eta(i)} \left(1 - \frac{f_1}{l_1(i)}\right).$$

Conditional on the employment size $l_1(i)$, revenue productivity is lower for producers with higher subsidy rates $\eta(i)$.

Higher subsidy rate promotes the producer to expand its production and employment. Facing downward sloping demand, the producer charges lower price. Since the physical productivity is unaffected by subsidy, however, lower price translates to lower revenue productivity. Therefore, variation in subsidy rates across producers encourages a negative association between revenue productivity, which is lower when $\eta(i)$ is high, and employment size, which is higher when $\eta(i)$ is high.

On the extensive margin, the entry condition (11) places a lower bound on the profit potential $h(i)$. Because employment is monotonically increasing in $h(i)$, this condition imposes a lower bound on employment size, similar to the case without subsidies. On the other hand, subsidy

rate heterogeneity relaxes the bounds on the sales-to-employment ratio. This is in contrast to the case with fixed overhead cost heterogeneity discussed in Section 3.4.3, which relaxes the lower bound on employment but maintains bounds on the sales-to-employment ratio. As a result, subsidy rate heterogeneity can more easily maintain a negative relationship conditional on active producers.

Figure 4(b) illustrates this idea with a simulated relationship between the domestic sales-to-employment ratio and domestic employment. Whereas the simulation for Figure 4(a) allows each producer to vary in productivity and fixed overhead costs, the simulation for Figure 4(b) lets producers vary in productivity and domestic subsidy rate. The parameters are again chosen so that 80 % of producers are active in the domestic market. Due to the one-to-one relationship between profit and employment, inactive producers are also the smallest in terms of employment size.

In contrast to the case with heterogeneous fixed costs, the selection effect imposes no bound on the sales-to-employment ratio, and its conditional mean decreases with employment size above the threshold. Indeed, the relationship becomes monotonically decreasing when inactive producers are excluded, similar to the actual pattern observed in the Chinese data.

4.3 Export intensity and size

For a producer active in the domestic market, its export-to-domestic sales ratio is

$$\sum_{j=2}^{N} \frac{r_j(i)}{r_1(i)} = \sum_{j=2}^{N} \frac{D_j}{D_1} \left(\frac{\mu_j}{\mu_1}\right)^{\rho} \left(\frac{1}{t_j \eta(i)}\right)^{\frac{\rho}{1-\rho}} \mathbb{1}(z(i) > a_j).$$

The domestic subsidy rate $\eta(i)$ and productivity $z(i)$ play distinct roles in determining the producer's export intensity. A higher subsidy rate, which applies only to domestic sales, increases the producer's domestic market production but not exports, reducing export intensity. Since a higher subsidy rate results in larger total employment size, subsidy rate heterogeneity induces a negative relationship between export intensity and total employment. On the other hand, only the productivity determines the number of export destinations, as the subsidy on domestic sales has no influence on the potential profit from exports. Since higher productivity leads to larger employment as well, the number of export destination rises with size.

With sufficiently large variation in subsidy rates across producers, the intensive margin effect dominates, and the conditional mean of export intensity decreases with producer size. As a result, the model explains how the export intensity decreases with size, as observed with the

Chinese data, even though larger exporters tend to ship to a greater number of destinations.

5 Quantifying the gains from trade

The data patterns observed in the Chinese manufacturing sector qualitatively fail to match the predicted patterns under an optimal allocation, instead aligning with the presence of producer-specific subsidies. In this section, I quantify the importance of accounting for such producer-level misallocation when estimating the welfare gains from trade liberalization.

Specifically, I calibrate two models using the same producer-level dataset. The first model, which I call the full model, generalizes the model introduced in Section 4 and allows destination- and producer-specific subsidy rates. The second model, which I call the baseline, excludes subsidies. Due to the CES utility assumption, the market equilibrium allocation of the baseline model coincides with the social planner's.

With each model, I consider an exogenous increase in foreign supply and demand, reflecting changes in factors such as variable trade costs. I calculate the resulting changes in welfare, measured by CES-aggregated consumption, to determine the size of the GFT. Applying this exercise to the Chinese manufacturing data reveals that ignoring producer-level misallocation leads to a significant overestimation of the GFT — Compared to the baseline model, the full model predicts 40 % smaller GFT.

Derivation details for this section are provided in Appendix C.

5.1 Setting

The estimation model considers a two-country setting, $N = 2$, reflecting the dataset which reports only the total export value for each producer without specifying exports by destination. For mnemonic purposes, I label country 1 as d for the domestic destination and country 2 as x for the export destination.

The full model generalizes the setting from Section 4.1 by allowing producer-specific subsidies on export sales. Thus, each producer i is endowed with three variables: productivity $z(i)$, the subsidy multiplier on domestic sales $\eta_d(i)$, and the subsidy multiplier on export sales $\eta_x(i)$. For each destination $j \in \{d, x\}$, producer i's profit potential is denoted by $h_j(i) := \eta_j(i)^{\frac{1}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}}$, and its sales potential by $k_j(i) := \eta_j(i)^{\frac{\rho}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}}$.

The distribution of productivity and subsidy rates determines the equilibrium outcomes solely through the distribution of profit and sales potentials, which, in turn, affects the equilibrium

through the following functions for each destination $j \in \{d, x\}$:

$$S_j(h^*) := \mathrm{E}[\mathbb{1}(h_j(i) \geq h^*)], \quad H_j(h^*) := \mathrm{E}[h_j(i)\mathbb{1}(h_j(i) \geq h^*)], \quad K_j(h^*) := \mathrm{E}[k_j(i)\mathbb{1}(h_j(i) \geq h^*)]. \tag{12}$$

The first function, $S_j(h^*)$, represents the share of producers with profit potential above the threshold h^*. The second function, $H_j(h^*)$, sums the profit potentials $h_j(i)$ above the cutoff h^*.[9] It forms the kernel of the average-to-minimum ratio,[10] which previous studies have identified as a key statistic in determining equilibrium outcomes under similar models (Bas et al., 2017; Fernandes et al., 2023). The last function, $K_j(h^*)$, sums the *sales potentials* $k_j(i)$ among producers whose *profit potential* $h_j(i)$ is above the cutoff h^*.

In addition to the productivity and subsidy rate distributions, the model takes the foreign demand shifter D_x and the import price index P_m as given.[11] As a result, the model can be estimated using a dataset from only the country of interest without requiring assumptions about the productivity distribution or the extent of misallocation in foreign countries.

For notational ease, I capture the accessibility of market j with

$$O_j := D_j^{\frac{1}{1-\rho}} t_j^{-\frac{\rho}{1-\rho}}.$$

Since both D_x and t_x are given, the foreign market accessibility O_x is also given. The domestic market accessibility, $O_d = D_d^{\frac{1}{1-\rho}}$, depends on endogenously determined consumer expenditure E and the price index P. A trade liberalization episode is represented by an increase in foreign market accessibility O_x, capturing a rise in export opportunities, and a decrease in the import price index, P_m, capturing a rise in import competition.

5.2 Equilibrium

The equilibrium conditions follow standard assumptions: the representative consumer maximizes utility, producers maximize profit, labor and variety markets clear, and trade is balanced.

The profit of producer i in destination j is determined by its corresponding profit potential $h_j(i)$. Consequently, there is a cutoff profit potential level h_j^* such that producers sell in destination j

[9] Because the mass of producers is normalized to unity, the expected value equals the total value.

[10] $\frac{\mathrm{E}[h_j(i)|h_j(i)\geq h^*]}{h^*} = \frac{H_j(h^*)}{S_j(h^*)h^*}$.

[11] Caliendo and Feenstra (2024) and Demidova et al. (2024) show these assumptions hold under a limiting case of multi-country models.

if and only if their profit potential exceeds this cutoff. Each cutoff level satisfies

$$h_j^* = \left(\frac{w}{\rho}\right)^{\frac{\rho}{1-\rho}} \left(\frac{w f_j}{(1-\rho) O_j}\right). \tag{13}$$

The labor market clearing condition requires that total labor demanded by producers equals the economy's labor endowment $\bar{L}$:

$$\bar{L} = \sum_{j \in \{d,x\}} \left(f_j S_j(h_j^*) + \left(\frac{\rho}{w}\right)^{\frac{1}{1-\rho}} O_j H_j(h_j^*) \right). \tag{14}$$

Under the goods markets clearing condition, the price index satisfies

$$P^{-\frac{\rho}{1-\rho}} = \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} K_d(h_d^*) + P_m^{-\frac{\rho}{1-\rho}}. \tag{15}$$

Lastly, trade is balanced under equilibrium. I assume all domestic producers are owned by the domestic representative consumer, so the aggregate profit, net of aggregate subsidies, adds to national income. Together with the labor market clearing condition (14), the balanced trade condition can be expressed as

$$E = \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} \sum_j O_j K_j(h_j^*). \tag{16}$$

An equilibrium is a tuple (h_d^*, h_x^*, w, P, E) such that equations (13), (14), (15), and (16) hold, taking as given the scalar parameters $(\rho, \bar{L}, f_d, f_x, O_x, P_m)$ and the functions (12).

The GFT is measured as an increase in aggregate consumption, $\log Q = \frac{1}{\rho} \log U = \log E - \log P$, resulting from higher foreign market accessibility $\log O_x$ or lower import price index $\log P_m$.

5.3 Calibration

5.3.1 Distributional functions

I propose a way to estimate the functions (12) without imposing a functional form assumption on the underlying joint distribution of productivity and subsidy rates.

Although profit and sales potentials are not directly observed, their distribution can be estimated from production employment $v_j(i) := l_j(i) - f_j \propto h_j(i)$ and sales $r_j(i) \propto k_j(i)$. Since the actual datasets contain only the total employment of each producer, $v_j(i)$ need to be inferred.

Working backward, suppose that $v_j(i)$ has been successfully inferred. As shown in Appendix C.3,

the equilibrium conditions can be reformulated with a change of variables that replaces $h_j(i)$ and $k_j(i)$ with $v_j(i)$ and $r_j(i)$. For example, the profit cutoff condition, $h_j(i) \geq h_j^*$, translates to

$$v_j(i) \geq v_j^* := \frac{\rho}{1-\rho} f_j. \tag{17}$$

Similarly, the functions S_j, H_j, and K_j from (12) translate to, respectively,

$$S_j^v(v^*) := \mathrm{E}[\mathbb{1}(v_j(i) \geq v^*)], \quad V_j(v^*) := \mathrm{E}[v_j(i)\mathbb{1}(v_j(i) \geq v^*)], \quad R_j(v^*) := \mathrm{E}[r_j(i)\mathbb{1}(v_j(i) \geq v^*)]. \tag{18}$$

Because $v_j(i)$ and $r_j(i)$ are proportional to $h_j(i)$ and $k_j(i)$, respectively, these functions preserve the elasticities of their counterparts.

Given a dataset containing $(r_j(i), v_j(i))_{j\in\{d,x\}}$, these functions can be estimated with the sample mean analogs.[12] For example, the functions for $j = d$ can be estimated by calculating, for each value of $v^* \geq 0$, the share of producers whose $v_d(i)$ exceeds v^* and the sum of their production employment $v_d(i)$ and sales $r_d(i)$.[13]

5.3.2 Destination-specific employment

To estimate the relevant functions with sample means, destination-specific production employment, $v_j(i)$, needs to be inferred for each producer i.

For a non-exporting producer i, the production employment is easily inferred given the value of the domestic fixed overhead cost f_d, with $v_d(i) = l(i) - f_d$ and $v_x(i) = 0$.

For an exporter i, its production employment ratio satisfies

$$\frac{v_x(i)}{v_d(i)} = \left(\frac{O_x}{O_d}\right)^{-\frac{1-\rho}{\rho}} \left(\frac{r_x(i)}{r_d(i)}\right)^{\frac{1}{\rho}}.$$

Together with the identity $l(i) = \sum_j \left(f_j + v_j(i)\right)$, its destination-specific production employments $(v_d(i), v_x(i))$ can be inferred from the dataset $(r_d(i), r_x(i), l(i))$ and the values of ρ, $\frac{O_x}{O_d}$, f_d, and f_x.

Following much of the literature, I take the value of the love-of-variety parameter ρ as given. For the main results, I use $\rho = \frac{2}{3}$, which implies an elasticity of substitution of 3, consistent with values used in Hsieh and Klenow (2009) and Bai et al. (2024).

[12] Technically, the sample means need to be scaled by the proportion of the survey population relative to all "potential" producers. However, this scaling factor does not influence the GFT estimation given that $S_j(h_j^*)$ is fixed during the estimation procedure as described below.

[13] In practice, I use a sigmoid function in place of the indicator function for smoothing. That is, in place of $\mathbb{1}(v_j(i) \geq v^*)$, I use $(1 + e^{-\alpha(v_j(i)-v^*)})^{-1}$, with the scale parameter set at $\alpha = 10$.

In the absence of subsidies, the market accessibility ratio $\frac{O_x}{O_d}$ would equal the aggregate export intensity among exporters. However, differences in "average" export and domestic subsidies introduce a wedge that may result in excessive or insufficient overall trade.[14] The extent of the aggregate distortion cannot be determined within the framework of the social planner's solution in Section 3, which is agnostic regarding the optimal level of total exports. To remain conservative, I assume that the observed level of aggregate export intensity among exporters is undistorted. Accordingly, the same accessibility ratio value is used in both the full and baseline models.

The overhead fixed costs f_j govern the extensive margin of producer activity. Therefore, the share of active producers in each market provides a natural target for estimating the fixed costs. One might consider all observed producers with positive sales as active. However, this approach would imply an implausibly low value for fixed costs, given that some observed producers have only a single employee. To address this issue without complicating the model with heterogeneous fixed costs or producer dynamics, I assume that some observed producers with positive sales are unprofitable and thus would be inactive under equilibrium.

For the main results, I assume that 80 % of the observed producers are active in the domestic market.[15] The fixed costs f_d and f_x are jointly estimated so that the sample analog of $S_d^{\nu}(\nu_d^*)$ equals 0.8 and $\frac{S_x^{\nu}(\nu_x^*)}{S_d^{\nu}(\nu_d^*)}$ matches the observed share of exporters, 17.1 %.[16] Given the ad hoc choice of the survival threshold, however, I check the sensitivity of the results to different target threshold values.

Equipped with the values of ρ, $\frac{O_x}{O_d}$, f_d, and f_x, $\nu_j(i)$ can be inferred from the dataset. In turn, the functions in (18) can be estimated using sample analogs.

The functions are estimated on a limited domain determined by the range of the observed values of $\nu_j(i)$. When considering the effect of expanded trade opportunities, which lower the profit potential cutoff for exports, the lower bound on the domain for foreign-destination functions poses an issue. Essentially, the counterfactual estimation requires inferring which producers would begin exporting as export opportunities expand.

This problem presents a challenge, as export profit potential cannot be inferred for non-exporting producers. While a non-exporter's sales and employment pin down its productivity $z(i)$ and

[14] The aggregate export intensity among exporters satisfies $\frac{R_x(\nu_x^*)}{R_d(\nu_x^*)} = \left(\frac{O_x}{O_d}\right) \frac{\mathrm{E}\left[\eta_x(i)^{\frac{\rho}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}} \mathbb{1}(h_x(i) \geq h_x^*)\right]}{\mathrm{E}\left[\eta_d(i)^{\frac{\rho}{1-\rho}} z(i)^{\frac{\rho}{1-\rho}} \mathbb{1}(h_x(i) \geq h_x^*)\right]}$.

[15] The 80 % threshold roughly matches the 2-year survival rate of manufacturing enterprises (OECD, 2013).

[16] Since the sample analog estimates the share of the survey population rather than the "true" population of all potential producers, the assumed value of active producers is actually less than 80 %, scaled down by the (unknown) survey-to-true population ratio.

domestic subsidy multiplier $\eta_d(i)$, the fact that it does not export provides only an upper bound, $\eta_x^m(i)$, on its export subsidy multiplier. Pinning down $\eta_x(i)$ would require knowing the counterfactual value of its exports.

To estimate the distribution of the export subsidy rate among non-exporters, I use the joint distribution estimated from exporters, conditioning on each non-exporter's observable characteristics. This process begins by estimating the joint density of $(z(i), \eta_d(i), \eta_x(i))$ for exporters using kernel density estimation. Then, for each non-exporter, I draw its export subsidy multiplier $\eta_x(i)$ from this estimated joint density, conditioned on its productivity, domestic subsidy rate, and the export subsidy bound $\eta_x(i) \leq \eta_x^m(i)$. With these draws in hand, I estimate the functions S_x, H_x, and K_x across all observed producers. The process is repeated 100 times, and the resulting functions are averaged pointwise.

Appendix C shows that ρ, f_d, f_x, and the functions (18) are sufficient to determine the GFT. In particular, the initial values of the trade parameters O_x and P_m do not affect the equilibrium response to changes in these parameters.

5.3.3 Baseline model

The baseline model is a special case of the full model with $\eta_d(i) = \eta_x(i) = 1$ for all producers i. As such, producers are distinguished only by their productivity $z(i)$, with no distinction between profit and sales potentials or between domestic and export markets. In other words, $h_d(i) = h_x(i) = k_d(i) = k_x(i)$ for all producers i. As a result, $S := S_d = S_x$ and $H := H_d = H_x = K_d = K_x$.

Each producer's productivity $z(i)$ can be inferred from its domestic sales $r_d(i)$, circumventing the need to infer production employment $v_j(i)$ for each destination. The equilibrium conditions are reformulated with a change of variables that replaces productivity $z(i)$ with domestic sales $r_d(i)$. For example, the productivity cutoffs are translated into domestic sales cutoffs: one for being profitable in the domestic market, r_{dd}^*, and another for the foreign market, r_{dx}^*.

With the change of variable, functions S and H translate to the following functions based on the domestic sales distribution:

$$S^r(r_d^*) := \mathrm{E}[\mathbb{1}(r_d(i) \geq r_d^*)], \quad R_d(r_d^*) := \mathrm{E}[r_d(i)\mathbb{1}(r_d(i) \geq r_d^*)]. \tag{19}$$

These functions are estimated with the sample analogs.

The baseline model takes the same love-of-variety parameter value ρ as the full model. The market accessibility ratio $\frac{O_x}{O_d}$ and the overhead fixed costs f_d and f_x are calibrated to match the same targets: the aggregate export intensity among exporters and the share of active producers

in each destination market.

5.4 Calibration results

I apply the estimation procedure to the Chinese firm-level data. Although the full model can account for pure exporters through high export subsidy rates, exporters with export intensity greater than 90 % are excluded to maintain consistency with earlier sections. To ensure robustness, the sensitivity analysis considers higher threshold values of 95 % and 99 %. Since the baseline model calibration excludes producers with no domestic sales, this exclusion also ensures that the same sample is used across models.

In the full model, with $\rho = \frac{2}{3}$, the estimated values of the fixed cost parameters are $\log(f_d) = 2.66$ and $\log(f_x) = 1.08$. Under the baseline model, the estimated parameter values are $\log(f_d) = 5.13$ and $\log(f_x) = 6.26$.

Appendix Figure A.3 plots the functional parameters (12) on a log-log scale for both the baseline and full models.[17] Subfigure A.3(a) plots the baseline survival function S, effectively revealing the domestic sales distribution. The shape of the domestic sales distribution in China appears similar to that observed in other developed countries,[18] underscoring the importance of examining the joint distribution. Subfigures A.3(c) and A.3(d) display the corresponding functions for the full model. In the absence of subsidies, $S_d = S_x$ and $H_d = H_x = K_d = K_x$. Therefore, differences in these functions drive the equilibrium effects of misallocation.

To further assess the model calibration, I present in Figure 5 the kernel density plots of four firm-level characteristics in the Chinese manufacturing sector: observed sales-to-employment ratio, inferred productivity $z(i)$, domestic subsidy rates $\eta_d(i)$, and export subsidy rates $\eta_x(i)$. Each subfigure displays the densities for four subgroups of firms based on export status and state ownership. The estimation procedure itself is blind to firm's ownership status. As such, the comparison of subsidy rate distributions between private and state-owned firms provides a useful check against the prior expectation that state-owned firms receive higher subsidy rates.

Subfigure 5(a) plots the densities of the sales-to-wage ratio across the four firm subgroups. As suggested earlier by Figure 2, state-owned firms exhibit a lower revenue productivity compared to private firms. Furthermore, as noted in Table 1, there is a lack of productivity premia for exporters, especially among private firms, when the output is measured with the shipment value.

[17] The translation vector between (18) and (12) consists of parameters that are not required for GFT estimations. For illustration purposes, arbitrary values were assigned to those parameters.

[18] For example, it can be directly compared to the domestic sales distribution of French firms shown in Eaton et al. (2011), Figure 2.

Subfigure 5(b) displays the densities of inferred "true" productivity for each subgroup. The calibrated model captures the exporter premium, consistent with the findings of Li et al. (2020) that Chinese exporters exhibit higher physical productivity. Meanwhile, the model also maintains that SOEs have lower productivity than private firms.

Subfigure 5(c) depicts the inferred density distributions of domestic subsidy rates for each group. Consistent with prior expectations, SOEs have higher average subsidy rates than private firms, reflecting their greater access to both direct and indirect subsidies from the state. Within each ownership group, exporters exhibit lower subsidy rates on domestic sales than non-exporters, aligning with the model prediction that domestic subsidies enable firms to be active without encouraging exports.

Lastly, Subfigure 5(d) shows the densities of export subsidy rates for exporters and the maximum possible export subsidy rates for non-exporters. State-owned firms continue to exhibit higher average export subsidy rates than their private counterparts. The dispersion of subsidy rates within each exporter group is also notable, comparable to the variation in productivity rates, reflecting the within-exporter misallocation discussed in Section 2.3. Moreover, the significant gap in subsidy rates between exporters and non-exporters points to substantial misallocation along the extensive margin, echoing the findings of Khandelwal et al. (2013) that the selection of exporters in China was suboptimal.

5.5 Gains from trade

I examine the effects of bilateral trade liberalization by considering a simultaneous increase in foreign market accessibility, O_x, and a decrease in the import price index, P_m. The change in log bilateral trade openness is defined as $\Delta \log T := \frac{1-\rho}{\rho}\Delta \log O_x = -\Delta \log P_m$. In other words, a unit increase in $\log T$ implies a $\frac{1-\rho}{\rho}$ increase in $\log O_x$ and a unit decrease in $\log P_m$. This scenario could result, for example, from simultaneous and proportionate decreases in variable export and import costs, holding the set of imported varieties fixed.[19]

Figure 6 shows the changes in key outcome variables in response to increases in bilateral trade openness under the baseline and full models. Subfigure 6(a) illustrates the GFT, showing how aggregate consumption responds to changes in bilateral trade openness. The results show that the baseline model significantly overestimates the size of the GFT compared to the full model. Specifically, under the baseline model, a percentage reduction in bilateral trade costs, or

[19] Since $\Delta \log O_x = \frac{1}{1-\rho}\Delta \log D_x - \frac{\rho}{1-\rho}\Delta \log t_x$, a unit decrease in $\log t_x$ leads to a $\frac{\rho}{1-\rho}$ increase in $\log O_x$. Moreover, since $-\frac{\rho}{1-\rho}\Delta \log P_m = -\frac{\rho}{1-\rho}\Delta \log t_m$ when $\Delta \log p(\omega) = \Delta \log t_m$ for all imported varieties ω, a unit decrease in $\log t_m$ results in a unit decrease in $\log P_m$. Together, these changes lead to a unit increase in $\log T$ as defined.

an equivalent change in foreign supply and demand, results in a 0.50 percent increase in real consumption. In contrast, the full model predicts an increase of only 0.29 percent from the same shock, about 40 % lower than the baseline estimation.

For comparison, Melitz and Redding (2015) use a model calibrated to the U.S. data and find that reducing variable trade costs from 1.8 to 1.0 increases welfare by 17 % under a Pareto-distributed productivity, compared to 12 % under a degenerate productivity distribution. These changes correspond to elasticity values of 0.27 and 0.19, respectively, implying approximately 30 % lower GFT when dropping producer heterogeneity. Similarly, Adão et al. (2020) find that allowing for non-constant elasticity increases the GFT estimate by around 10 % for China compared to the model presuming constant elasticity.[20] These comparisons suggest that accounting for market distortions, at least in the context of China, has greater quantitative implications than differences in assumptions about the productivity distribution.

Bai et al. (2024) also account for domestic misallocation in their model and estimate its impact on GFT using Chinese data. Assuming a joint lognormal distribution for productivity and subsidies, they find that moving to autarky would *improve* China's welfare by 3.7 %. My non-parametric approach similarly indicates that market distortions reduce the GFT but predicts it remains positive. This discrepancy highlights the importance of distributional assumptions in the context of market distortions, as they can lead to qualitatively different conclusions about whether misallocation enhances, dampens, or reverses the GFT.

Subfigure 6(b) illustrates the change in total export value. Despite the substantial difference in GFT estimates, both models predict similar trade elasticities, approximately 1.7. This result underscores the limitations of the sufficient statistics approach in the context of developing countries, as models with similar trade elasticities can yield vastly different GFT predictions.

Subfigures 6(c) and 6(d) shed light on the GFT gap between the two models. Subfigure 6(c) illustrates the response of the share of exporting producers to trade liberalization, revealing a stark contrast: the baseline model predicts a much larger increase in the share of exporters compared to the full model. Compared to the baseline model, the full model attributes a significantly larger role to incumbents in driving export growth. As shown in Subfigure 6(d), the reduced reallocation along the extensive margin in the full model results in lower growth in aggregate productivity defined as the CES-averaged productivity conditional on being active.[21]

The results remain robust to ad hoc choices made during the estimation process. Appendix

[20] Melitz and Redding (2015) and Adão et al. (2020) use different definitions of GFT: the former measures real consumption changes in response to trade costs, while the latter focuses on responses to trade share.

[21] Specifically, aggregate productivity is defined as $\left(\mathrm{E}[z(i)^{\frac{\rho}{1-\rho}} \mid h(i) \geq h_d^*]\right)^{\frac{1-\rho}{\rho}}$.

Figure A.4 presents the GFT outcomes of repeating the same exercise under varying assumptions for the love-of-variety parameter ρ, the target share of active producers, and the threshold export intensity used for sample inclusion.

For the love-of-variety parameter ρ, I consider values of $\rho = \frac{3}{4}$ and $\rho = \frac{4}{5}$, corresponding to elasticity of substitution values of 4 and 5, respectively. Higher values result in larger GFT under both models, though the baseline model continues to substantially overestimate the gains.

In the main result, I set the target share of active producers, $S_d(h_d^*)$, to 80 %. Lowering or raising it by 10 percentage point makes negligible differences in GFT estimates for both the baseline and full models.

Lastly, I check the sensitivity of the results to the sampling choice. In the main analysis, exporters with export intensity exceeding 90 % are excluded. Increasing this threshold to 95 % or 99 % results in slightly higher GFT estimates for both models, though the baseline model continues to predict significantly larger gains than the full model.

6 Conclusion

In a broad class of models with heterogeneous producers, revenue productivity and export intensity are expected to increase with size under an efficient allocation. Thus, examining the relationship between these variables and producer size offers a lens to detect producer-level misallocation in an economy. In contrast to the patterns observed in the U.S., both revenue productivity and export intensity decrease with producer size in the Chinese manufacturing sector These findings align with prior research showing significantly greater producer-level misallocation in China compared to the U.S. Moreover, the observed patterns indicate that misallocation in China extends beyond the state-owned sector, as the negative size-dependence persists even among private firms.

Accounting for producer-level misallocation is crucial for estimating the gains from trade liberalization. When calibrated to Chinese manufacturing data from 2000, a model that ignores producer-level misallocation significantly overestimates the size of the GFT. The findings suggest that China's "reform and opening-up" policies worked synergistically.

Without privatization efforts that mitigated state-induced domestic market distortions, its integration into the global economy may have been less successful. More broadly, market distortions may offer a quantitatively significant explanation for why some countries fail to realize the expected benefits from trade liberalization. The main results rely on just three variables per producer — domestic and export sales and total wages — making these methods readily applicable

to other countries.

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A Additional figures and tables

B Theoretical derivations

Proof of Lemma 1. Problem (3) can be solved in two steps. First, the planner determines the set of active producers A_j. Then for each active producer $i \in A_j$, she determines the (positive) output quantity $q_j(i)$.

Consider the second stage problem, where the set of active producers A_j has already been determined. The first order condition directly implies (4), where $\mu_j = \left(\frac{\rho}{\lambda_j}\right)^{\frac{1}{1-\rho}}$, and λ_j is the Lagrangian multiplier associated with destination j labor constraint.

It remains to determine the optimal set of active producers, A_j. Substituting the expression for the optimal quantity (4) into the labor constraint in (3) yields an expression for the multiplier:

$$\mu_j = V_j(A_j)^{\rho-1} Z_j(A_j)^{1-\rho}, \quad V_j(A_j) := \bar{L}_j - \int_{A_j} f_j \, \mathrm{d}i, \quad Z_j(A_j) := \int_{A_j} \tilde{z}_j(i) \, \mathrm{d}i.$$

Substituting this expression for the multiplier back into the output quantity (4) and then into the objective function (3) yields the maximal utility value in country j from consuming country 1's varieties as a function of A_j:

$$U_j(A_j) = V_j(A_j)^{\rho} Z_j(A_j)^{1-\rho}. \tag{20}$$

In the first stage problem, the planner maximizes $U_j(A_j)$ by choosing A_j. Since $V_j(A_j)$ depends on A_j through and only through its measure, holding the share of producers active in market j fixed, the planner chooses the producers to maximize $Z_j(A_j)$.

Clearly, this is achieved by choosing the producers with the highest productivity $z(i)$. In other words, the optimal set of producers selling to market j can be characterized as $A_j = \{x \in [0,1] \mid z(i) \geq a_j\}$ for some productivity cutoff value $a_j \geq 0$, chosen such that $1 - G(a_j)$ meets the target share of exporting producers.

Proof of Claim 1. The shipment value to total employment ratio of non-exporting producers can be decomposed as

$$\frac{r(i)}{l(i)} = \frac{p_1(q_1(i))q_1(i)}{l_1(i)} = \left(\frac{p_1(q_1(i))q_1(i)}{l_1(i) - f_1}\right)\left(\frac{l_1(i) - f_1}{l_1(i)}\right).$$

By Lemma 2, under the optimal allocation, the expression simplifies to

$$\frac{r(i)}{l(i)} = D_1 \mu_1^{\rho-1}\left(1 - \frac{f_1}{l_1(i)}\right),$$

which is increasing in non-exporting producer i's total employment $l(i) = l_1(i)$.

Proof of Claim 2. I show that export to total shipment ratio of exporter i is non-decreasing in $z(i)$, while total employment is strictly increasing in $z(i)$.

The export to total shipment ratio, $\frac{\sum_{j=2}^N r_j(i)}{\sum_{j=1}^N r_j(i)} = \frac{\sum_{j=2}^N r_j(i)}{r_1+\sum_{j=2}^N r_j(i)}$ is non-decreasing in $z(i)$ if and only if the export to domestic shipment ratio, $\frac{\sum_{j=2}^N r_j(i)}{r_1(i)} = \sum_{j=2}^N \frac{r_j(i)}{r_1(i)}$, is non-decreasing in $z(i)$. By Lemma 2,

$$\sum_{j=2}^N \frac{r_j(i)}{r_1(i)} = \sum_{j=2}^N \frac{D_j}{D_1}\left(\frac{\mu_j}{\mu_1}\right)^\rho \left(\frac{t_1}{t_j}\right)^{\frac{\rho}{1-\rho}} \mathbb{1}(z(i) > a_j).$$

The constant term of each summand on the right-hand-side of the equation is positive and therefore non-decreasing in $z(i)$ with discrete jumps at each a_j.

The total employment of firm i can be expressed as

$$\sum_{j=1}^N l_j(q_j(i);x) = \sum_{j=1}^N (f_j + \mu_j \tilde{z}_j(i))\mathbb{1}(z(i) > a_j),$$

which is strictly increasing in $z(i)$.

VES utility. Suppose the representative consumer's utility is described by the VES function (8). The inverse demand function takes the form

$$p_j(q) = D_j u'(q) \tag{21}$$

where $D_j > 0$ is the destination j demand shifter.

The first order condition of the planner's problem becomes

$$u'(q_j(i)) = \mu_j l_j'(q_j(i);x)$$

where $\mu_j > 0$ is the multiplier associated with labor constraint. Since $l_j(q_j(i);x) - f_j = l_j'(q_j(i);x)q_j(i)$,

$$\frac{p_j(q_j(i))q_j(i)}{l_j(q_j(i);x) - f_j} = \frac{\mu_j}{D_j}.$$

Claim 1 follows.

Define the elasticity of marginal utility $\beta(q) = -\frac{qu''(q)}{u'(q)}$. Under the optimal allocation, the sales

ratio of firm i takes the form

$$\frac{r_j(i)}{r_1(i)} = \left(\frac{D_j}{D_1}\right)\frac{u'(q_j(i))q_j(i)}{u'(q_1(i))q_1(i)} = \left(\frac{D_j\mu_j t_j}{D_1\mu_1 t_1}\right)\frac{q_j(i)}{q_1(i)}.$$

The idiosyncratic component $\frac{q_j(i)}{q_1(i)}$ is increasing in $z(i)$ if and only if $\beta(q_j) \leq \beta(q_1)$ since the planner's first order condition requires

$$q_j(i) = (u')^{-1}(\mu_j t_j z(i)^{-1}).$$

Therefore, Claim 2 holds if the parameters are such that $q_j \leq q_1$ and $\beta(q)$ is increasing.

Heterogeneous variable costs. Suppose that for each producer i, the variable export cost to destination $j \geq 2$ is given by $t_j(i) = t_j\tilde{t}(i)$.

Claim 1 is unaffected. By Lemma 2, the variation of sales-to-employment ratio among non-exporting producers is solely driven by the variation in $\tilde{z}_1(i)$, which in turn depends solely on $z(i)$.

To see how heterogeneous trade cost affects Claim 2, I separately consider the effect on the extensive and intensive margins.

On the extensive margin, larger exporters still export to greater number of destinations. To see why, note that Lemma 1 still holds with redefining of $\tilde{z}_j(i) := \left(\frac{z(i)}{t_j(i)}\right)^{\frac{\rho}{1-\rho}} = \left(\frac{1}{t_j}\tilde{z}(i)\right)^{\frac{\rho}{1-\rho}}$, where $\tilde{z}(i) := \frac{z(i)}{\tilde{t}(i)}$. As a result, the set of producers exporting to market j is now characterized by $A_j = \{i \in [0,1] \mid \tilde{z}_j(i) > a_j\}$. In other words, producers with the highest trade-cost adjusted productivity are assigned to ship to foreign destinations. At the same time, $\tilde{z}(i)$ determines producer i's labor allocation: $l_j(i) = f_j + \mu_j\tilde{z}(i)$. Therefore, producers with higher $\tilde{z}(i)$ export to greater number of destinations and employ more labor, leading to a positive relation between the two.

On the intensive margin, the equalization result no longer holds. For a producer i that sells to market j, the shipment value ratio $\frac{r_j(i)}{r_1(i)}$ is decreasing in $t_j(i)$. At the same time, conditional on productivity $z(i)$, a producer with higher trade cost $t_j(i)$ has lower employment size. This observation suggests that under a "typical" joint distribution of $z(i)$ and $\tilde{t}(i)$, there should be a positive relationship between export intensity and size on the intensive margin as well.

To better characterize such distribution, consider the expected value of export intensity $\frac{r_j(i)}{r_1(i)}$ conditional on producer size $l(i)$. Since the employment size is pinned down by the adjusted productivity $\tilde{z}(i)$, it is sufficient to consider $\mathrm{E}[\log\big(r_j(i)\big) - \log(r_1(i)) \mid \log(\tilde{z}(i))] = \alpha - \frac{\rho}{1-\rho}\mathrm{E}[\log\tilde{t}(i) \mid \log(\tilde{z}(i))]$, where α is some constant. Therefore, a sufficient condition for the export intensity

to be non-decreasing in size is that the mean trade cost conditional on adjusted productivity is non-increasing, i.e. $\mathrm{E}[\log \tilde{t}(i) \mid \log \tilde{z}(i)]$ is non-increasing in $\log \tilde{z}(i)$.

Since $\mathrm{Cov}[\log \tilde{t}(i), \log \tilde{z}(i)] = \mathrm{Cov}[\log \tilde{t}(i), \log z(i)] - \mathrm{Var}[\log \tilde{t}(i)]$, a sufficient condition for this is $\mathrm{Cov}[\log \tilde{t}(i), \log z(i)] \leq \mathrm{Var}[\log \tilde{t}(i)]$, assuming finite second moments.

Heterogeneous fixed costs. Suppose that for each producer i, the fixed cost associated with sales to destination $j \geq 1$ is given by $f_j(i) = f_j \tilde{f}(i)$.

Consider the planner's problem. From equation (20), the marginal change of U_j due to the inclusion of producer i in A_j equals

$$\mathrm{d} \log U_j(A_j) = \frac{(1-\rho)\tilde{z}_j(i)}{Z_j(A_j)} - \frac{\rho f_j(i)}{V_j(A_j)},$$

which is positive if and only if

$$\frac{\tilde{z}_j(i)}{f_j(i)} > \frac{\rho}{1-\rho} \frac{Z_j(A_j)}{V_j(A_j)}.$$

Therefore, producer i sells in destination j if and only if $a_j(i) := \tilde{z}_j(i) f_j(i)^{-1}$ is above a threshold value, say $a_j > 0$.

From a straightforward extension of Lemma 2, producer i's sales-to-employment ratio associated with the domestic market is

$$\frac{r_1(i)}{l_1(i)} = D_1 \mu_1^{\rho-1} \frac{\mu_1 a_1(i)}{1 + \mu_1 a_1(i)},$$

which is increasing in $a_1(i)$. Given $a_1(i) \geq a_1 > 0$ from the selection, the sales-to-employment ratio of producer i conditional on $i \in A_1$ is is bounded below by a positive number. Furthermore, even without the selection, the ratio is also bounded above by $D_1 \mu_1^{\rho-1}$. Therefore, even if the distribution of $f(i)$ is unbounded, the conditional distribution of log sales-to-employment ratio is bounded, as claimed in the text.

C Quantitative model details

This appendix provides details on the misallocation model used for the counterfactual exercises.

C.1 Model description

The representative consumer's maximization of utility (1) implies the following inverse demand for a variety

$$p_d(q) = E^{1-\rho} P^{\rho} q^{-(1-\rho)}$$

where

$$P^{-\frac{\rho}{1-\rho}} := \int_{\omega\in\Omega} p(\omega)^{-\frac{\rho}{1-\rho}}\,\mathrm{d}\omega$$

is the CES price index and E is the total expenditure. Define domestic demand shifter $D_d := E^{1-\rho}P^{\rho}$. The foreign inverse demand follows (5) with the foreign demand shifter $D_x > 0$ exogenously given.

Producer i maximizes profit for each destination $j \in \{d, x\}$:

$$\max_{q} \quad \eta_j(i)p_j(q)q - wl_j(q;i)$$

where the labor requirement $l_j(q;i)$ is given in (2). The producer outcomes at the interior solution are

$$\begin{aligned}
p_j^*(i) &= \left(\frac{wt_j}{\rho}\right)k_j(i)^{-\frac{1-\rho}{\rho}},\\
r_j^*(i) &= \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j k_j(i),\\
l_j^*(i) &= f_j + \left(\frac{\rho}{w}\right)^{\frac{1}{1-\rho}} O_j h_j(i),\\
\pi_j^*(i) &= (1-\rho)\left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j h_j(i) - wf_j.
\end{aligned} \tag{22}$$

C.2 Equilibrium

Extensive margin cutoffs. Producer i sells in market j if and only if $\pi_j^*(i) \geq 0$. That is,

$$q_j^*(i) \geq 0 \iff h_j(i) \geq h_j^*$$

where h_j^* is given in equation (13).

Labor market clearing. The total labor demand equals $\sum_j \mathrm{E}[l_j^*(i)\mathbb{1}(h_j(i) \geq h_j^*)]$ where $l_j^*(i)$ is given in (22). Condition (14) follows by equating the labor supply $\bar{L}$ with the demand.

Price index. The representative consumer has access to domestic varieties Ω_d and imported varieties Ω_m which form a partition of all available varieties Ω. Therefore, the CES price index P satisfies

$$P^{-\frac{\rho}{1-\rho}} = \int_{\omega\in\Omega_d} p(\omega)^{-\frac{\rho}{1-\rho}}\,\mathrm{d}\omega + \int_{\omega\in\Omega_m} p(\omega)^{-\frac{\rho}{1-\rho}}\,\mathrm{d}\omega\,.$$

The second summand, the price index of imported varieties, is given exogenously and denoted P_m, so that $P_m^{-\frac{\rho}{1-\rho}} := \int_{\omega\in\Omega_m} p(\omega)^{-\frac{\rho}{1-\rho}}\,\mathrm{d}\omega$.

Under the profit maximization, the first summand, the price index of domestic varieties, equals $\mathrm{E}\left[p_d^*(i)^{-\frac{\rho}{1-\rho}}\mathbb{1}(h_d(i)\geq h_d^*)\right]$ where $p_d^*(i)$ is given in (22). Condition (15) follows.

Balanced trade. All domestic producers are owned by the representative consumer, and all subsidies and taxes are transferred from the consumer. Consequently, the income equals the sum of labor income and the aggregate profit minus the aggregate net subsidies. The labor income equals to $w\bar{L}$, and the aggregate profit equals

$$\Pi := \sum_j \mathrm{E}[\pi_j^*(i)\mathbb{1}_{h_j(i)\geq h_j^*}] = \sum_j \left((1-\rho)\left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j H_j(h_j^*) - w f_j S_j(h_j^*)\right).$$

The value of subsidy received by producer i for sales in market j equals $(\eta_j(i)-1)r_j(i)$. (If $\eta_j(i)<1$, then producer i is taxed the absolute value.) The net subsidy aggregated across all active producers equals

$$N := \sum_j \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j \left(H_j(h_j^*) - K_j(h_j^*)\right).$$

Combined, the balanced trade condition requires $E = w\bar{L} + \Pi - N$, or

$$E = w\left(\bar{L} - \sum_j f_j S_j(h_j^*)\right) + \sum_j \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j \left(K_j(h_j^*) - \rho H_j(h_j^*)\right).$$

Combined with the labor market clearing condition (14), it simplifies to the expression (16).

Trade liberalization. For each equilibrium conditions (13), (14), (15), and (16), I take a log transformation and derivative with respect to $\log O_x$ and $\log P_m$ to derive the following system of linear equations:

$$\begin{aligned}
\mathrm{d}\log h_d^* &= \frac{1}{1-\rho}\mathrm{d}\log w - \frac{\rho}{1-\rho}\mathrm{d}\log P - \mathrm{d}\log E,\\
\mathrm{d}\log h_x^* &= \frac{1}{1-\rho}\mathrm{d}\log w - \mathrm{d}\log O_x,\\
\frac{\rho}{1-\rho}\mathrm{d}\log P &= \gamma_d\left(\frac{\rho}{1-\rho}\mathrm{d}\log w - \varepsilon_d^K(h_d^*)\mathrm{d}\log h_d^*\right) + \gamma_x\left(\frac{\rho}{1-\rho}\mathrm{d}\log P_m\right)\\
0 &= \sum_j \gamma_j^L\left(\gamma_j^F\varepsilon_j^S(h_j^*)\mathrm{d}\log h_j^* + (1-\gamma_j^F)\left(\mathrm{d}\log O_j - \frac{1}{1-\rho}\mathrm{d}\log w + \varepsilon_j^H(h_j^*)\mathrm{d}\log h_j^*\right)\right)\\
\mathrm{d}\log E &= -\frac{\rho}{1-\rho}\mathrm{d}\log w + \sum_j \gamma_j\left(\mathrm{d}\log O_j + \varepsilon_j^K(h_j^*)\mathrm{d}\log h_j^*\right)
\end{aligned} \tag{23}$$

where $\gamma_j := \frac{O_j K_j(h_j^*)}{\sum_j O_j K_j}$ is the market j share of sales, which under the balanced trade also equals the

expenditure share, $\gamma_j^L := \frac{L_j}{\bar{L}}$ is market j share of total employment, $\gamma_j^F = \frac{f_j S_j(h_j^*)}{L_j}$ is the fixed cost share of total cost associated with market j, and $\varepsilon_j^F(h^*)$ is the elasticity of function F_j evaluated at h^*.

C.3 Change of variables

To facilitate parameter estimations from the observables, I consider the following changes of variables:

$$v_j(h_j) := \left(\frac{\rho}{w}\right)^{\frac{1}{1-\rho}} O_j h_j, \quad r_j(k_j) := \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_j k_j.$$

The empirical functions (18) relate to the parameter functions (12) by

$$S_j^v(v_j(h^*)) = S_j(h^*), \quad V_j(v_j(h^*)) = v_j(H_j(h^*)), \quad R_j(v_j(h^*)) = r_j(K_j(h^*)).$$

Since v_j and r_j are linear transformations, the corresponding function pairs have the same elasticity.

The profit potential cutoff (13) translates to production employment cutoff $v_j^* = v_j(h_j^*)$ given in (17). The conditions for labor market clearing (14), price index (15), and balanced trade (16) satisfy

$$\begin{aligned} \bar{L} &= \sum_j \left(f_j S_j^v(v_j^*) + V_j(v_j^*) \right), \\ \frac{R_d(v_d^*)}{E} &= 1 - \left(\frac{P_m}{P}\right)^{-\frac{\rho}{1-\rho}}, \\ E &= R_d(v_d^*) + R_x(v_x^*). \end{aligned}$$

The coefficients in the system (23) are determined by ρ, f_d, f_x, and the functions (18). Specifically, $\gamma_j = \frac{R_j(v_j^*)}{\sum_k R_k(v_k^*)}$ and $L_j = f_j S_j^v(v_j^*) + V_j(v_j^*)$, where v_j^* is determined by ρ and f_j given by (17). Furthermore, as discussed in the main text, the elasticities of the functions (12) evaluated at h_j^* equal the elasticities of the estimated functions (18) evaluated at v_j^*.

C.4 Baseline model

In the absence of misallocation, $\eta_d(i) = \eta_x(i) = 1$ for all producers i. Consequently, $h_d(i) = h_x(i) = k_d(i) = k_x(i) = z(i)^{\frac{\rho}{1-\rho}} =: h(i)$ for all producers i. These equalities imply that the equilibrium can be characterized with two functions, $S(h^*) := \mathrm{E}[\mathbb{1}(h(i) > h^*)]$ and $H(h^*) := \mathrm{E}[h(i)\mathbb{1}(h(i) > h^*)]$.

The domestic sales $r_d(i)$ is proportional to $h(i)$:

$$r_d(i) := p_d^*(i) q_d^*(i) = \left(\frac{\rho}{w}\right)^{\frac{\rho}{1-\rho}} O_d h(i).$$

With the change of variable, the functions S and H have analogs S^r and R_d defined in (19).

The domestic sales at the cutoff productivity levels h_d^* and h_x^* satisfy

$$r_{dd}^* = \frac{w f_d}{1-\rho}, \quad r_{dx}^* = \frac{O_d}{O_x}\left(\frac{w f_x}{1-\rho}\right).$$

The price index satisfies

$$\left(1 - \frac{R_d(r_{dd}^*)}{E}\right) P^{-\frac{\rho}{1-\rho}} = P_m^{-\frac{\rho}{1-\rho}}$$

where $R_d(r_{dd}^*)$ is the aggregate domestic sales under the equilibrium.

The labor market clearing condition is

$$\bar{L} = f_d S_d + f_x S_x + \left(\frac{\rho}{w}\right) R_d(r_{dd}^*) + \left(\frac{\rho}{w}\right)\left(\frac{O_x}{O_d}\right) R_d(r_{dx}^*)$$

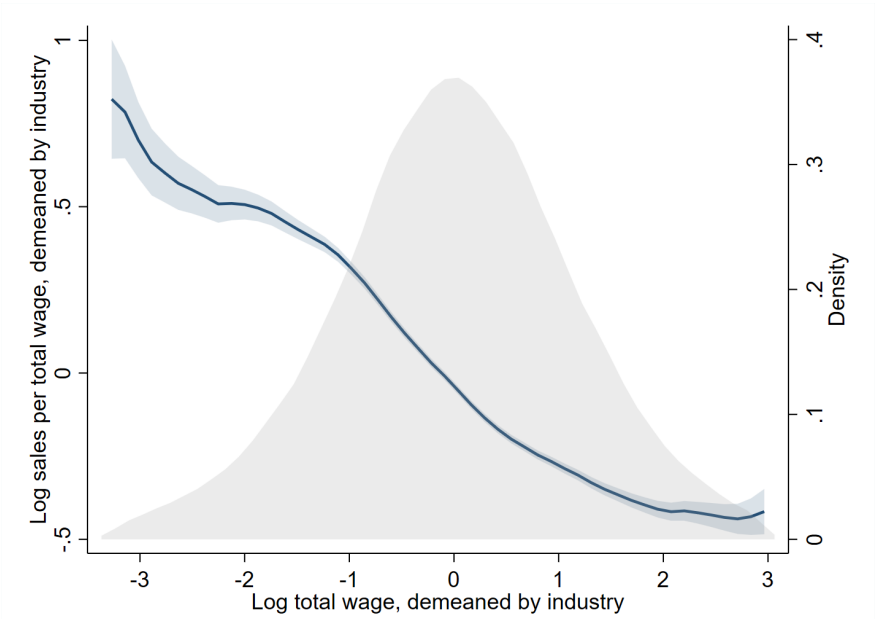
where $R_d(r_{dx}^*)$ captures the the aggregate domestic sales of exporters.

Lastly, the balanced trade condition simplifies to

$$E = w\left(\bar{L} - f_d S^r(r_{dd}^*) - f_x S^r(r_{dx}^*)\right) + (1-\rho) R_d(r_{dd}^*) + (1-\rho)\left(\frac{O_x}{O_d}\right) R_d(r_{dx}^*) = R_d(r_{dd}^*) + \left(\frac{O_x}{O_d}\right) R_d(r_{dx}^*),$$

where the second equality uses the labor market clearing. With the balanced condition, the labor market clearing condition further simplifies to

$$w\left(\bar{L} - f_d S^r(r_{dd}^*) - f_x S^r(r_{dx}^*)\right) = \rho E.$$



(a) China



(b) U.S.

Figure 1: Revenue productivity and producer size

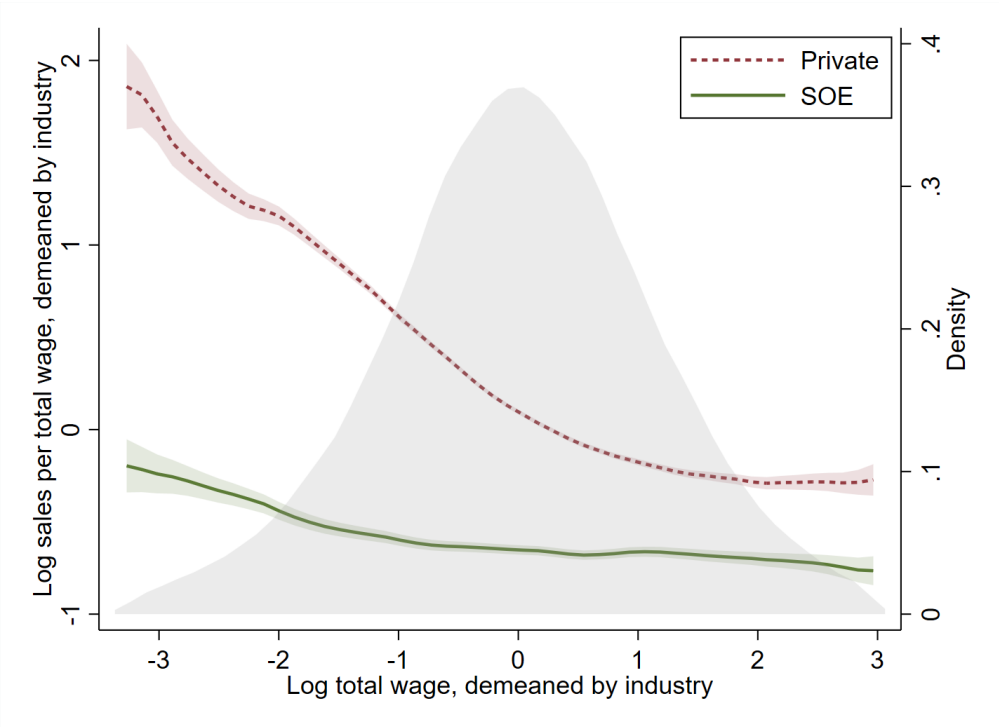
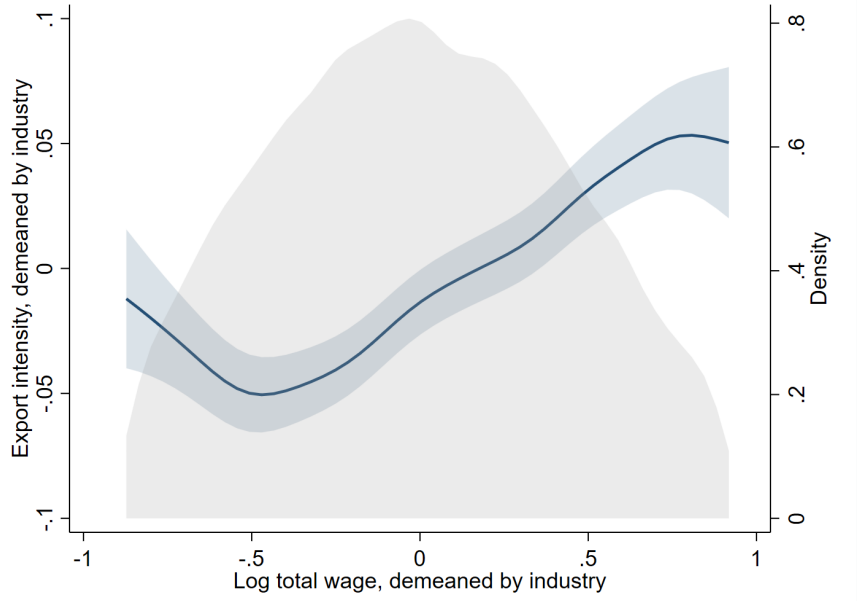


Figure 2: Revenue productivity and size in China: Role of SOEs

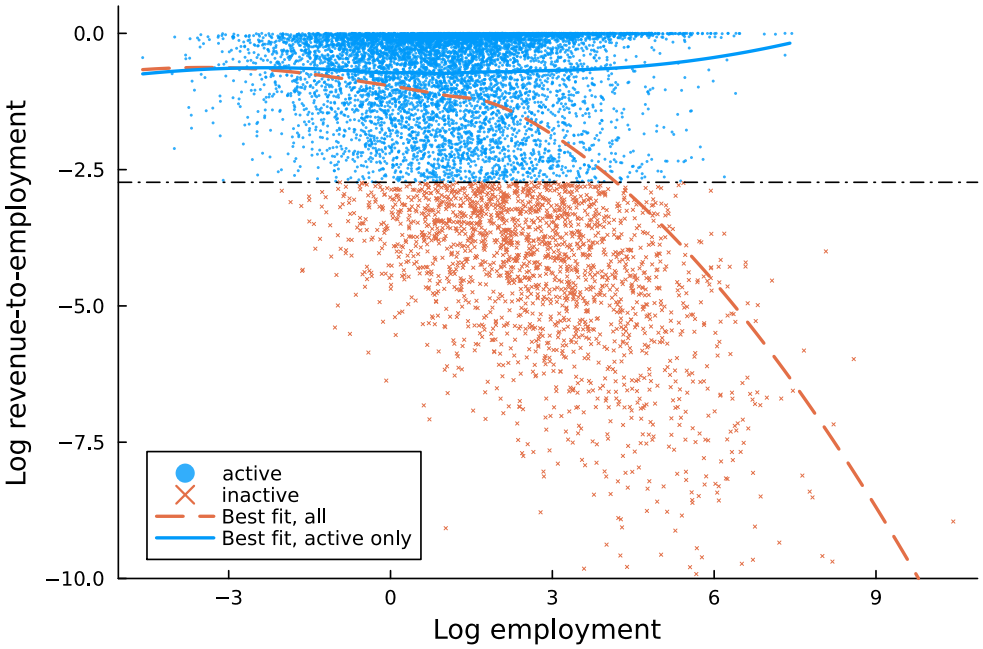


(a) China

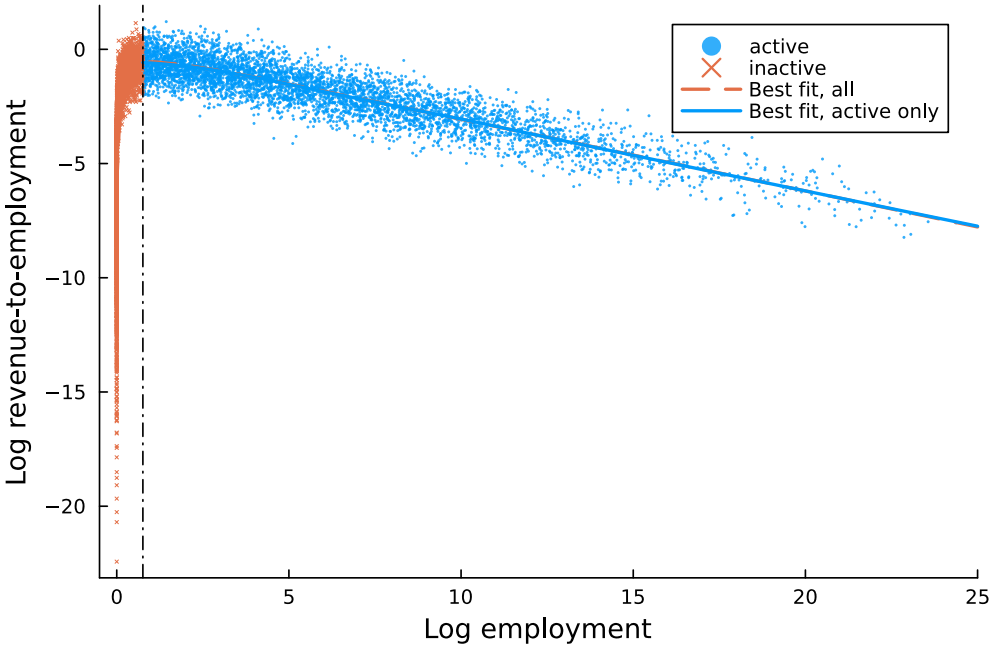


(b) U.S.

Figure 3: Export intensity and producer size

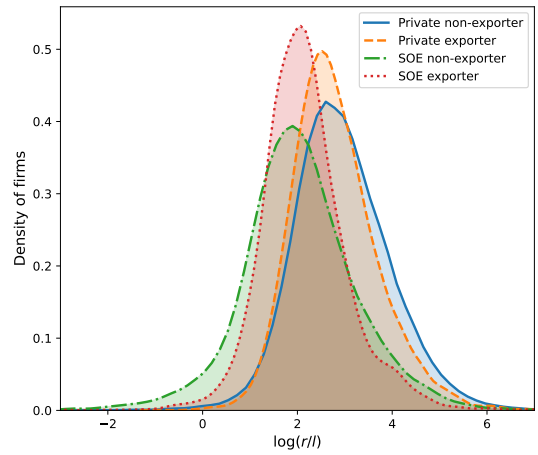


(a) Heterogeneous fixed costs

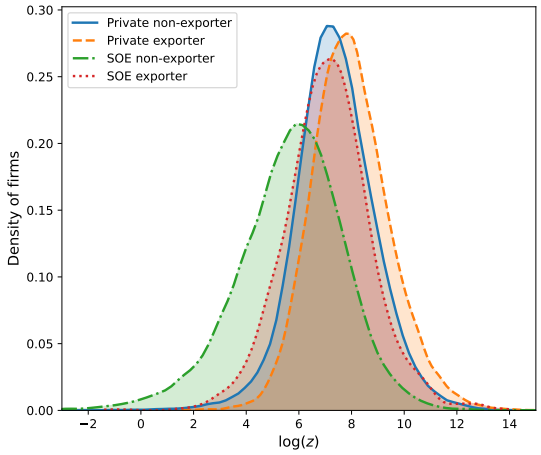


(b) Heterogeneous subsidy rates

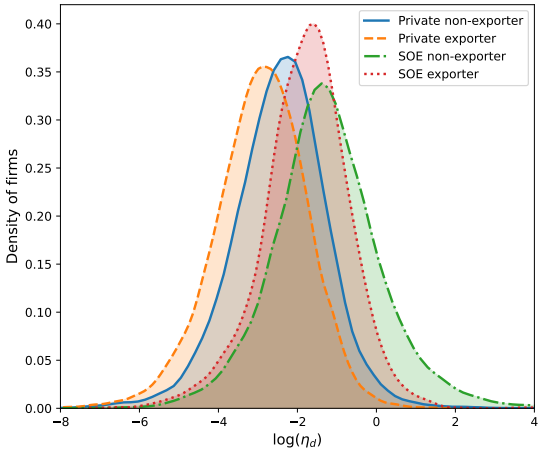
Figure 4: Revenue productivity and producer size simulations



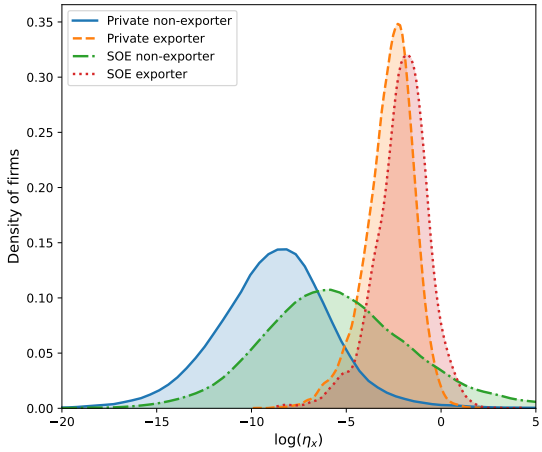
(a) Sales-to-wage ratio



(b) Productivity

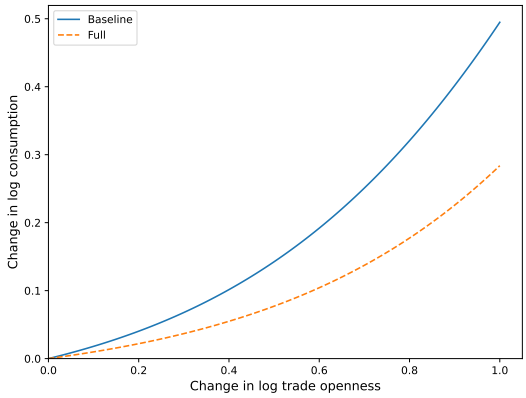


(c) Domestic subsidy rate

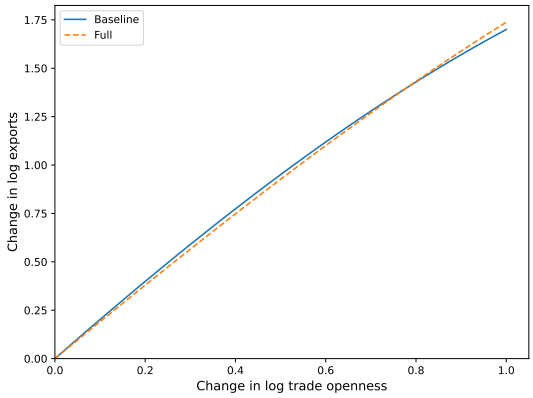


(d) Export subsidy rate

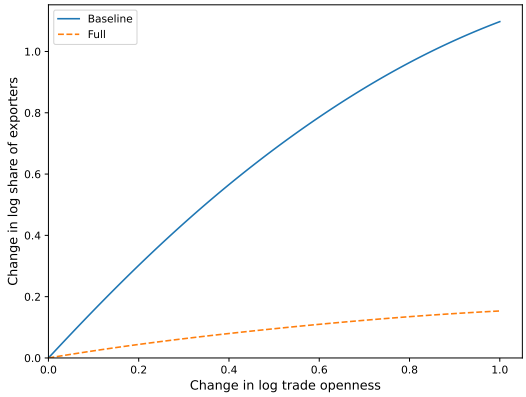
Figure 5: Distributions of firm productivity and subsidy rates in Chinese manufacturing



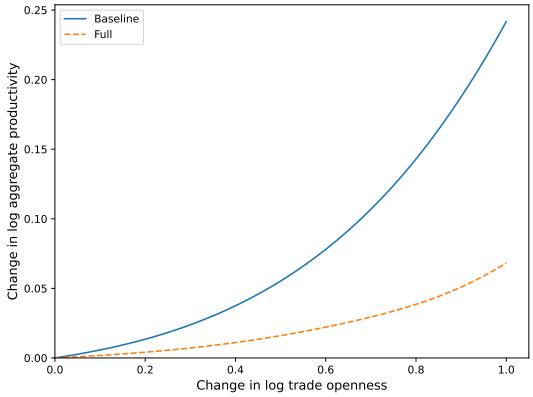
(a) Real consumption



(b) Total exports

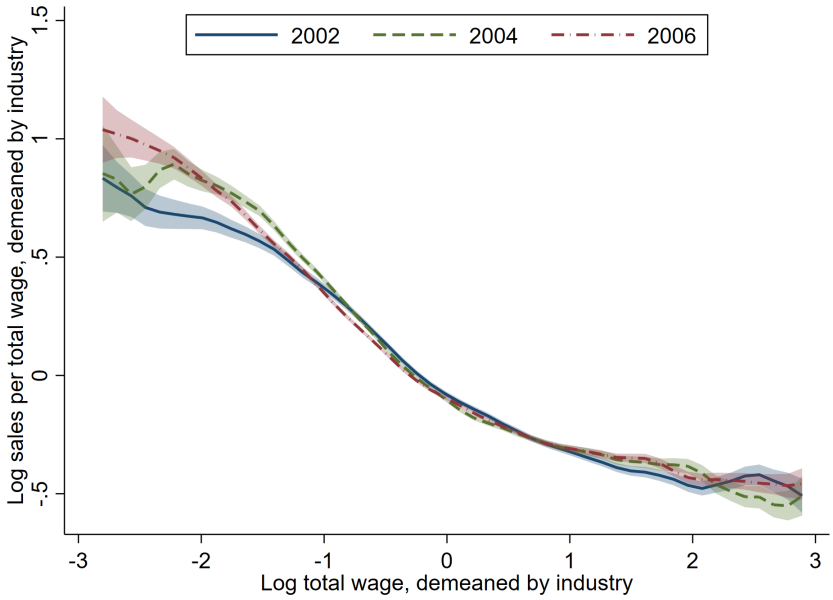


(c) Share of exporters

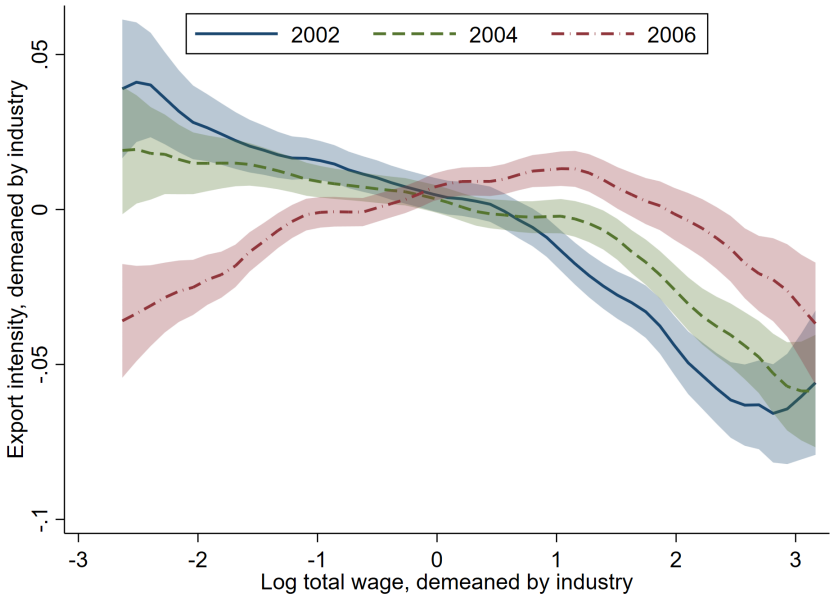


(d) Aggregate productivity

Figure 6: Effects of trade liberalization under two models

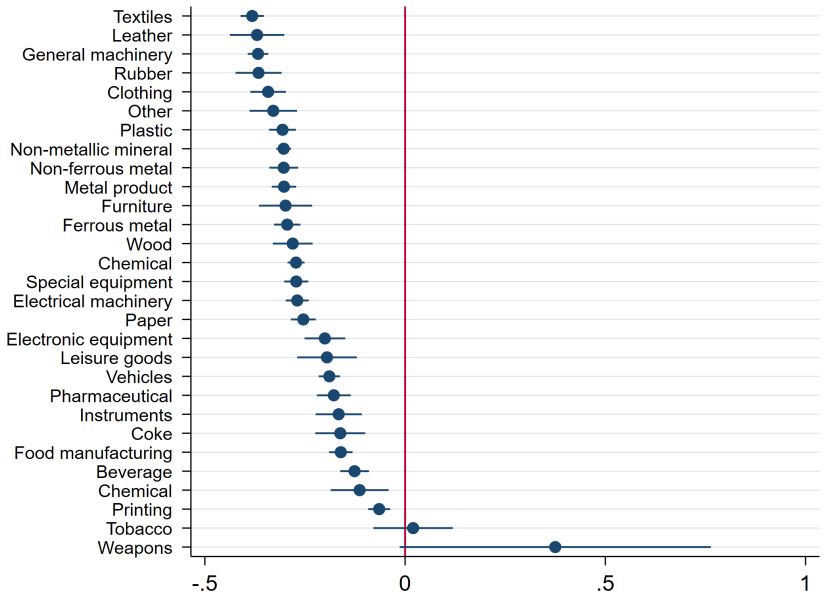


(a) Revenue productivity and size

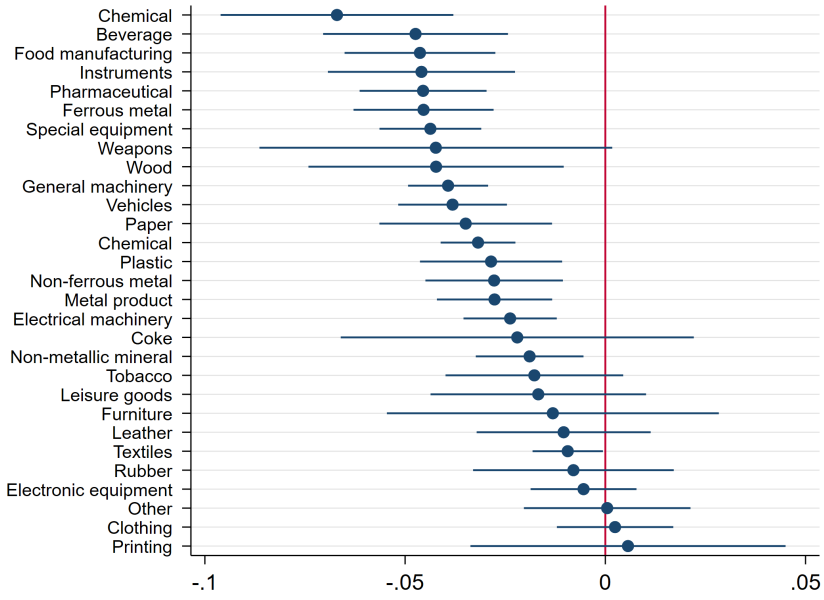


(b) Export intensity and size

Figure A.1: Chinese manufacturing patterns across years

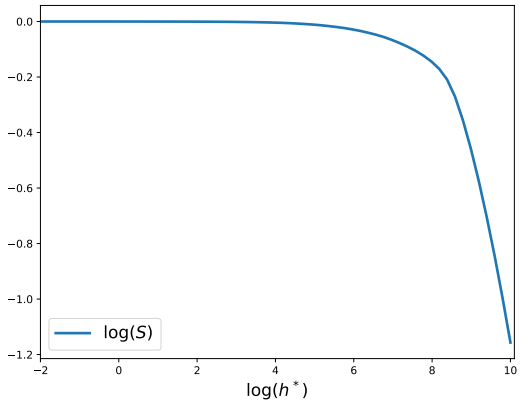


(a) Revenue productivity and size

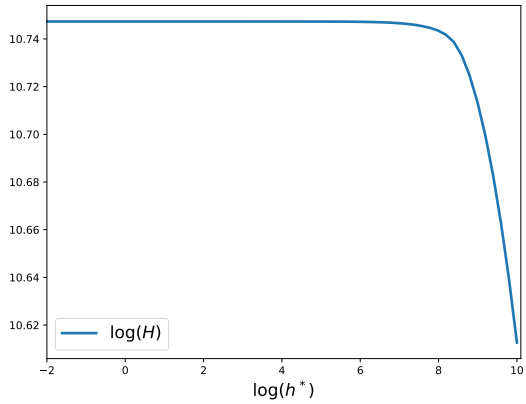


(b) Export intensity and size

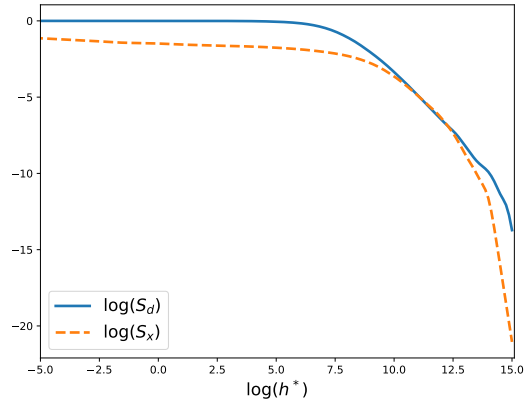
Figure A.2: Chinese manufacturing patterns by sector



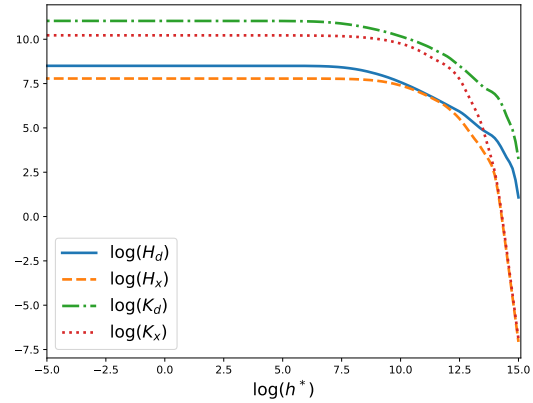
(a) Survival function (baseline)



(b) Potential function (baseline)

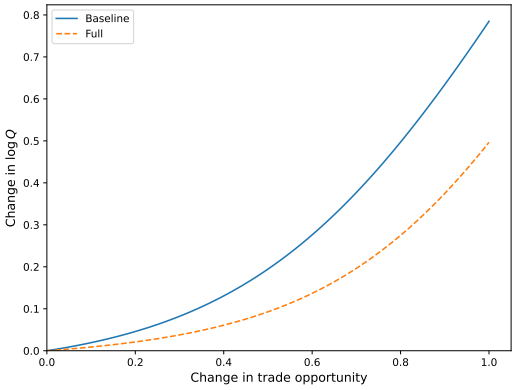


(c) Survival functions (full)

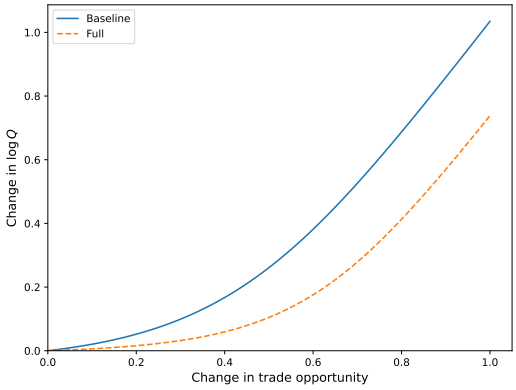


(d) Potential functions (full)

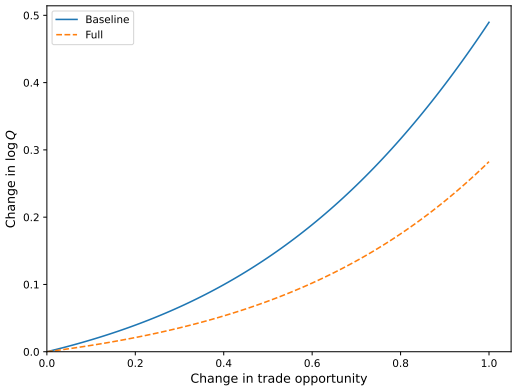
Figure A.3: Estimated function parameters on the Chinese data



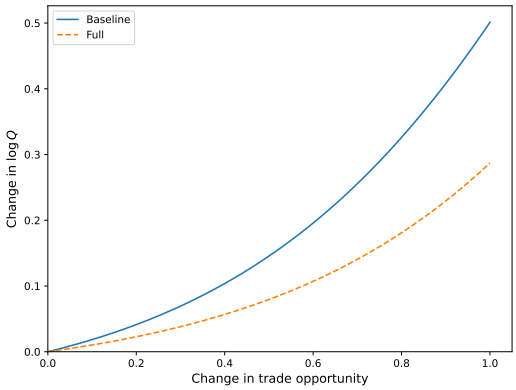
(a) $\rho = \frac{3}{4}$ ($\sigma = 4$)



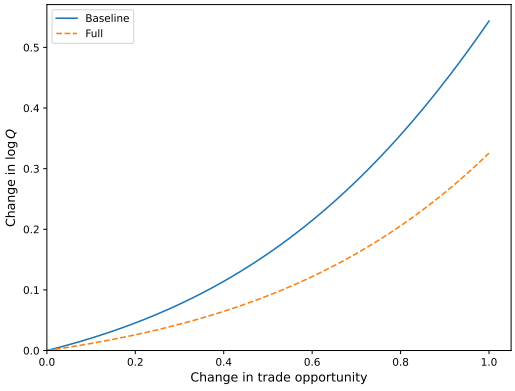
(b) $\rho = \frac{4}{5}$ ($\sigma = 5$)



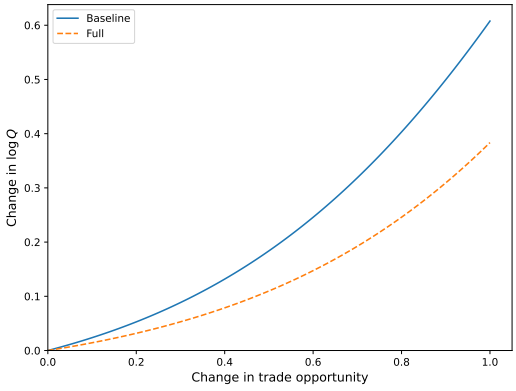
(c) Target $S_d = 0.7$



(d) Target $S_d = 0.9$



(e) Export intensity threshold at 95 %



(f) Export intensity threshold at 99 %

Figure A.4: Gains from trade estimation under alternate parameterization